Current Topics in Medicinal Chemistry, 2022, *22*, 1530-1552

RESEARCH ARTICLE

Anxiolytic and Antidepressant-like Effects of Monoterpene Tetrahydrolinalool and *In silico* Approach of new Potential Targets

Pablo Rayff da Silva[1], Natalia Diniz NunesPazos[1], Thallita Karla Silva do Nascimento Gonzaga[1], Jéssica Cabral de Andrade[1], Álefe Brito Monteiro[1], Anne Caroline Ribeiro Portela[1], Hugo Fernandes Oliveira Pires[1], Mayara dos Santos Maia[2], Diogo Vilar da Fonsêca[3], Marcus T. Scotti[2], José Maria Barbosa Filho[4], Damião Pergentino de Sousa[4], Cícero Francisco Bezerra Felipe[1], Reinaldo Nóbrega de Almeida[1] and Luciana Scotti[2,*]

[1]Psychopharmacology Laboratory, Institute of Drugs and Medicines Research, Federal University of Paraíba, 58051-085, Via Ipê Amarelo, S/N, João Pessoa, Paraíba, Brazil; [2]Cheminformatics Laboratory, Institute of Drugs and Medicines Research, Federal University of Paraíba, 58051-900, Via Ipê Amarelo, S/N, João Pessoa, Paraíba, Brazil; [3]Collegiate of Medicine, Federal University of São Francisco Valley, 48607-190, Rua Aurora, S/N, Bahia, Brazil; [4]Pharmaceutical Chemistry Laboratory, Department of Pharmaceutical Sciences, Federal University of Paraíba, 58051-900, Via Ipê Amarelo, S/N, João Pessoa, Brazil

ARTICLE HISTORY

Received: December 01, 2021
Revised: February 03, 2022
Accepted: February 16, 2022

DOI:
10.2174/1568026622666220505104726

Abstract: ***Introduction***: Although drugs currently available for the treatment of anxiety and depression act through modulation of the neurotransmission systems involved in the neurobiology of the disorder, yet they often present side effects, which can impair patient adherence to treatment.

Methods: This has driven the search for new molecules with anxiolytic and antidepressant potential. Aromatic plants are rich in essential oils, and their chemical constituents, such as monoterpenes, are being studied for these disorders. This study aims to evaluate the anxiolytic and antidepressant-like potential of the monoterpene tetrahydrolinalool in *in vivo* animal models and review pharmacological targets with validation through molecular docking. Male Swiss mice (*Mus musculus*) were treated with THL (37.5-600 mg kg^{-1} p.o.) and submitted to the elevated plus maze, open field, rotarod, and forced swim tests. In the elevated plus-maze, THL at doses of 37.5 and 75 mg kg^{-1} induced a significant increase in the percentage of entries (72.7 and 64.3% respectively), and lengths of stay (80.3 and 76.8% respectively) in the open arms tests.

Results: These doses did not compromise locomotor activity or motor coordination in the animals. In the open field, rotarod tests, and the forced swimming model, treatment with THL significantly reduced immobility times at doses of 150, 300, and 600 mg kg^{-1}, and by respective percentages of 69.3, 60.9 and 68.7%.

Conclusion: In molecular docking assay, which investigated potential targets, THL presented satisfactory energy values for: nNOs, SGC, IL-6, 5-HT_{1A}, NMDAr, and D1. These demonstrate the potential of THL (a derivative of natural origin) in *in vivo* and *in silico* models, making it a drug candidate.

Keywords: Tetrahydrolinalool, Natural products, Anxiety, Depression, Docking, Chemoinformatics.

1. INTRODUCTION

According to the World Health Organization, it is estimated that 28.8% and 16.6% of the population will respectively present either anxiety or depression during their lives. In 40-75% of cases, these disorders may appear in association with each other [1]. Patients presenting both anxiety and depression are typically less productive, making these diseases a principal cause of psychosocial disability. This is due to episodic aggravations, longer crises periods, accentuated symptoms, standard treatment refractoriness and ineffectiveness [2, 3]. Pharmacological therapy for anxiety generally involves the use of benzodiazepines, azapirone (buspirone), and antidepressant medications, among others. To treat depression, drugs from the class of tricyclics (TCAs), monoamine oxidase inhibitors (iMAOs), selective serotonin reuptake inhibitors (SSRIs), serotonin-noradrenaline reuptake inhibitors (SNRIs), or atypical antidepressants are used.

*Address correspondence to this author at the Cheminformatics Laboratory, Institute of Drugs and Medicines Research, Federal University of Paraíba, 58051-900, *via* Ipê Amarelo, S/N, João Pessoa, Paraíba, Brazil; E-mail: luciana.scotti@gmail.com

1873-4294/22 $65.00+.00

These drugs, although effective, have limitations due to their adverse effects, which include cognitive and motor alterations, as well as greater susceptibility of individuals to become tolerant and dependent [4, 5].

Considering the deficiencies of conventional therapy, aromatic plants provide an important pharmacological arsenal of alternatives for psychosomatic illnesses, such as anxiety and depression. Essential oils (EOs) are complex mixtures of secondary metabolites. Because of their high lipophilicity and volatility, they can permeate the blood-brain barrier, pharmacologically modulating various signaling pathways in the central nervous system (CSN) [6, 7].

The chemical constitution of EOs includes terpenic substances, formed by the so-called isoprene unit of five carbon atom subunits. Terpene compounds may structurally present ten (monoterpenes) or fifteen (sesquiterpenes) carbon atoms. Their pharmacological effects for treating neuropsychiatric disorders are widely described in the literature, and their study as both natural and intermediate compounds becomes more interesting with time, as we search for new drugs with both efficacy and better safety profiles [8, 9].

Plant species that produce linalool-rich EOs are used in traditional medicine for their sedative, anxiolytic, and antidepressant effects. Linalool is a tertiary alcohol (classified as a monoterpene) with central nervous system activity [10, 11]. Researchers have demonstrated that essential oils from the leaves of *Aniba rosaeodora* (Rosewood), *Aniba parviflora* (Macacaporanga), and *Aeollanthus suaveolens* (Catinga-de-mulata) present significant antidepressant activity (possibly due to the presence of linalool in their compositions), yet (in rodents) presenting no indications of spontaneous motor impairment or memory retention [10].

Previous studies examined the anxiolytic activity of linalool, revealing promising effects in classical behavioral models through mechanisms that involve GABAergic neurotransmission [12]. The anxiolytic-like effect of *Cinnamomum osmophloeum* leaf essential oil, rich in linalool, is well studied [13] and reductions in neurotransmitter concentrations (serotonin, dopamine, and noradrenaline) have been evidenced.

In such *in vivo* models, both the S (-) and S (+) linalool enantiomers present anxiolytic properties, without side effects. Given the direction and extent of studies with its derivatives, linalool has been revealed as a pharmacologically promising agent for anxiolytic and antidepressant response modulation. Tetrahydrolinalool (THL) (3,7-dimethyloctan-3-ol) is a cyclic alcohol monoterpene, produced during linalool metabolism, or obtained synthetically through an endocyclic double bond hydrogenation reaction (Fig. **1**) [14]. Due to its pleasant fragrance, it is used in cosmetic products, yet mentions of its pharmacological activities are practically non-existent in the literature.

The present study sought to evaluate the anxiolytic and antidepressant-like potential of the monoterpene tetrahydrolinalool *in vivo*, perform a brief review of important new targets for its respective activities, and validate these findings through computational prospecting (molecular docking) using the Molegro software Docker.

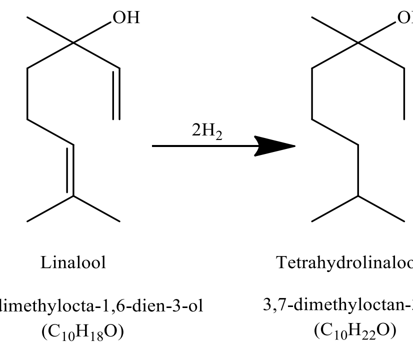


Fig. (1). Obtaining tetrahydrolinalool from linalool hydrogenation.

1.1. Neural Mechanisms for Anxiety and Depression

1.1.1. Monoaminergic Route - Receptor 5 Hydroxytryptamine 1A, 2A, and 2C (PDB: 6A93, 6BQH)

Serotonin, scientifically known as 5-hydroxytryptamine (5-HT), is a monoamine class neurotransmitter, synthesized by raphe nucleus neurons (RNN) (in the midbrain) from the amino acid L-tryptophan. Its production is not restricted to the CNS, and if deregulated, it will involve several mechanisms. 5-HT is closely related to psychiatric illnesses such as depression and anxiety. This is because the molecule is involved in the neurophysiology of many circuits, including emotions, memory, sleep, and thermal regulation [15]. According to its pharmacological functions and transduction mechanism, the 5-HT receptor family can be classified as composed of 14 subtypes such as 5-HT_{2A}, 5-HT_{2B}, and 5-HT_{2C} [16]. The 5-HT_{1A} isoform is one of the most abundant in the brain and can be found in two distinct populations: somatodendritic auto-receptors in raphe nuclei, and post-synaptic hetero-receptors - found in diverse central areas innervated by serotoninergic projections, in pyramidal neurons, and GABAergic interneurons [17-19].

5-HT receptors are classified as metabotropic, coupled to the Gi proteins, which cause depletion in cyclic adenosine monophosphate (cAMP) levels, inactivation of calcium channels, and activation of potassium channels; culminating in inhibition of neuronal activity. Auto-receptors, decrease the firing rates through negative feedback that limits firing release [20-22]. Pharmacological modulation of these receptors in raphe 5-HT nucleus neurons can minimize anxiety and depression behavior [23, 24], and partial 5-HT_{1A} receptor agonists such as buspirone are thus used to treat anxiety [25, 26].

Studies have shown that linalool exhibits antidepressant activity in the forced swimming model, the effect of which, in part, involves the monoaminergic neurotransmission, *via* 5-HT_{1A} receptors [27]. Psychopharmacological investigations of *Citrus aurantium* L. EO demonstrated anxiolytic activity, through receptors 5-HT_{1A}R. The test performed was the light-dark box test, employing the 5-HT_{1A} antagonist, WAY 100654. *Citrus aurantium* L. EO fruit has a wide chemical diversity, including the monoterpenes linalool, limonene, and α-terpineol, which may be responsible for its pharmacological effects.

As previously mentioned, 5-HT_{2A} receptors present activation similar to 5-HT_{1A}, however their mechanism promotes anxiogenic effects, explained in part, by an increase in glutamatergic response, *via* NMDA through Src kinase, which promotes increased neuronal excitation [28-30]. Models submitting mice to post-traumatic stress disorder have revealed that 5-HT_{2A} receptors play an important role in anxiety-like behavior - by inhibiting 5-HT_{1A} receptor expression through phosphorylation of ERK1 and ERK2 [31]. It was also observed that in endotoxic shock in rodents (induced by lipopolysaccharides) 5-HT_{2A} receptors elicit the signaling pathway mediated by mitogen-activated protein kinase (MAPK), inducing anxiety-like behavior [31].

Knowledge of new pharmacological agents of natural and synthetic origin, and with anxiolytic and antidepressant therapeutic effect has been extensively disseminated. Research involving molecular docking studies can reveal the structural bases for interaction patterns between amino acids (and interferences) when coupling. Molecular modeling with 8-aceti-7-hydroxy-4-methylcoumarin derivatives has demonstrated the importance of stabilizing the bond at 5-HT_{1A} and Asp155 (5-HT_{2A}) sites *via* a salt bridge formed between the basic piperazine nitrogen group and the amino acid aspartate 116 (Asp116). Such results allow targeting the most promising compounds using sub-nanomolar binding affinities [32].

The 5-HT_{1B} receptor group belongs to the GPCR family, with 43% 5-HT_{1A}R sequence homology, and yet differing in terms of cell location (mostly found in axon terminals), and in CNS distribution (greater basal ganglia density) [33]. Its functionality, like other 5-HT receptors, is to control extracellular serotonin levels, and consequently decrease neuronal firing rates. It is also known that these receptors present an additional mechanism in regulating reuptake transporters and/or influencing GABAergic (neocortex), cholinergic (prefrontal cortex and hippocampus), and dopaminergic (Nigro-striatal) neurotransmission [29, 33-36]. Evidence validating this receptor as a potential target in treating anxiety and depression is promising. However, it is still not clear how best to alter 5-HT_{1B}R binding (and thus activity) for optimal effect. Currently, the approved drug vortioxetine acts as a partial agonist of these receptors [33].

1.1.2. Soluble Guanylate Cyclase/Neuronal Nitric Oxide Synthase (SGC/nNOs) Pathway (PDB: 3UVJ and 5UO1)

Activation of sGC and nNOS enzymes is considered an important molecular pathway (for worsening moods) in patients with behavioral disorders. In animal stress models, elevated levels of nNOS and sGC have been found in the hypothalamus, hippocampus, and prefrontal cortex [37-39]. In mammals, there are three functional isoforms of nitric oxide synthase (NOS): nNOS (neuronal), eNOS (endothelial), and iNOS (inducible) [40]. Thus, nNOS was initially described in neurons, however it can be found in skeletal striated musculature, bronchial epithelium, and trachea; nNOS is regulated by calcium levels (Ca^{2+}). When stimulated, this isoform produces nitric oxide (NO) from L-arginine, in sufficient concentrations [40]. However, if produced in excess, NO is associated with worsening anxiety and depression, since it stimulates oxidative stress and neuroinflammation [41-44].

As a pro-inflammatory mediator NO is a key factor in the initiation of events that increase neuroinflammatory activity, principally, by promoting dilation and increased permeability of the blood-brain barrier, which favors migration of peripheral inflammatory cells to brain tissue [45]. Neuronal cells, in stressful situations, generate NO and superoxide anion (O_2^-) which can then join together to form the peroxynitrite anion ($ONOO^-$). Peroxynitrite is a powerful oxidant capable of initiating lipid peroxidation of the plasma membrane, damage DNA directly, cause protein dysfunction, and deplete fat-soluble antioxidants [46-48]. NO itself can also activate type 1 ryanodine receptors in the endoplasmic reticulum and increase concentration of cytosolic Ca^{2+} which, among other effects results in the generation of more free radicals and increased glutamate exocytosis, with consequent excitotoxicity mediated by excessive activation of NMDA receptors [49]. In postsynaptic neurons, hyperactive NMDA receptors cause abnormal Ca^{2+} influx, overstimulation of nNOS, and increased NO synthesis.

NO has been reported to target SGC, the enzyme responsible for converting guanosine 5-triphosphate (GTP) into cyclic guanosine 3,5-monophosphate (cGMP) [48], and the L-arginine-NO-cGMP pathway (modulated by nNOS and SGC) is an important signaling pathway. It is likely involved in regulating a variety of behavioral, cognitive, and emotional processes, including anxiety and depression [53-55]. Studies indicate that nNOS and SGC (inhibition) are valuable targets, and provide effective strategies for therapy of these behavioral disorders [50-52].

A computational molecular docking assay investigated the binding affinity and antidepressant activity of N'-methylene esters of omega-Nitro-L-arginine (L-Name), L-Nitro Arginine (7NA), 7-nitroindazole (7NI), and methylene blue with nNOs, and it was observed that coupling is favored through an inhibitor-target hydrogen bond, which can form stable complexes in close equilibrium states [53]. It was observed that the use of selective nNO inhibitors, such as 7NI, resulted in antidepressant effects in mice. This was evidenced by a decrease in immobility when evaluated during the tail suspension test. A protective effect on hippocampal serotoninergic terminations was also observed. The results validate the pharmacological potential of nNO inhibitors as new agents with anxiolytic and antidepressant potential [54].

As a possible mechanism for monoterpene antinociceptive activity, linalool inhibits the formation of NO and causes NMDA receptor blockade [55], and it has been suggested that its precursor (tetrahydrolinalool) acts by specifically inhibiting nNOs, thus modulating neural mechanisms related to anxiety and depression pathophysiology.

1.1.3. Inflammatory Pathway - TNF-α and IL-6 Cytokines (PDB: 2AZ5 and 1QE6)

In recent decades, it has been shown in patients with depression and anxiety disorders that there are direct correlations between immune dysfunction and increased inflammatory mediator levels; such as tumor necrosis factor alpha (TNF-α), interleukin 1beta (IL-1β), and interleukin-6 (IL-6) [56]. Cytokines are proteins or polypeptides capable of modulating inflammatory and immunological conditions. Upon reaching the CNS, they participate in the neurophysiological regulation of neural plasticity, and of circuits that involve neurotransmitter, and hormone metabolism [57].

Serotonin reuptake inhibitors (SSRIs), and tricyclic antidepressants reduce pro-inflammatory cytokine levels while increasing anti-inflammatory cytokine levels (such as IL-10). It is postulated that this is due to increased serotonergic receptor activation in immune cells, which controls the dynamics of pro- and anti-inflammatory cytokine release. Another mechanism seems to involve increased cAMP production, which activates protein kinase A (PKA) by increasing the production of the cAMP response element binding protein (CREB). This decreases the production of pro-inflammatory cytokines [58-60].

TNF-α is a pleiotropic cytokine with an important role in regulating cognitive functions in both anxiety and depression disorders [61, 62]. Its biological effects are exerted during TNFR1 and TNFR2 receptor binding, and expressed in both physiological and induced conditions. TNF-α activates the hypothalamic pituitary adrenal axis (HPA); increasing neuronal 5-HT transporter activity, and stimulating indoleamine 2,3-dioxygenase to decrease the concentration of tryptophan, and to thus impact the pathophysiology of anxiety and depression [61-64]. The molecule is also involved in activating brain microglia, as observed through synaptic plasticity marker reductions, and increasing neurodegenerative events [58].

In vivo research in db/db mice has indicated that blocking the TNF-α TLR-4/NF-κB signaling pathway, with agents such as etanercept, limits inflammatory conditions, and improves depression and anxiety-like behaviors. The same is found for anti-TNF-α monoclonal antibodies such as infliximab, which were shown to decrease immobility in the forced swimming test, anhedonia in the sucrose preference test, and anxiety in the elevated plus-maze (EPM) [58, 65, 66].

Central or peripheral administration of lipopolysaccharide (endotoxins derived from gram negative bacteria) promotes inflammation, increasing the release of inflammatory cytokines such as IL-6 and IL-1β. Immune system and brain communication dysfunctions can be related to CNS disturbances as mediated by cytokines [67]. In rodents, increases in IL-6 and IL-1β can influence mood, with manifestations similar to anxiety and depression, including anhedonia, decreased exploratory capacity, social withdrawal, and memory effects, among others. The exact mechanism is still poorly understood, but it is believed that alteration of serotonin metabolism, neural plasticity, and the HPA axis all influence long-term potentiation (LTP) [68]. Recently, additional mechanisms of cytokine interference have been discovered in the endocannabinoid system (CB1) which contribute to anxiogenic and depressive effects through GABA synapses in the striatum [69].

1.1.4. Glutamatergic Pathway - NMDA Receptor (PDB: 4PE5)

Despite the abundance of catecholamine neurotransmitters found in the neurochemistry of anxiety and depression, new therapeutic treatment approaches that act through different neural circuits are interesting [70]. Modulation of the glutamatergic pathway is a target of studies on anxiolytics and antidepressants. Glutamate is an excitatory neurotransmitter that acts on ionotropic and metabotropic receptors. The N-methyl-D-aspartate (NMDA) receptor, named after its selective agonist, is an ionic receptor controlled by ligands, a conductor of calcium current, found in high density in mesocorticolimbic regions, which mediates physiological responses related to synaptic function, plasticity, learning, memory, depression, anxiety, and fear [71]. Its crystallographic structure is composed of heterotetramers of the subunits: GluN1 and GluN2A/B/C/D, or GluN3A/B. The receptor’s carboxyterminal and aminoterminal subunit domains are important regions for regulatory activity [72, 73].

Preclinical studies have shown that NMDA receptor antagonists such as ketamine and analogs (phencyclidine and dizocilpine) act by decreasing the channel's activity, reducing Ca^{2+} influx, and mediating currents [73, 74]. In subanesthetic doses, ketamine acts in the pathophysiology of psychiatric disorders, reaching brain tissue in micromolar concentrations [74].

In summary, agents that block the NMDA receptor after the dissociation of glutamate in regions such as the ventral hippocampus and prefrontal cortex can affect emotional behavior. Studies with knockout mice have shown that deletion of receptor subunits and intra-hippocampal drug administration can improve anxiety and depression [75].The neurobiology of these disorders (neural mechanisms) is still obscure, yet, clinical data suggest that such antagonists may be related to the restoration of maladapted brain plasticity through inhibition of NMDA receptors in GABAergic interneurons in the medial prefrontal cortex, resulting in disinhibition of glutamatergic pyramidal neurons, and glutamatergic peak firing [76]. Blockade of NMDA receptors by antagonists such as memantine can induce greater BDNF mRNA and its trkB receptor expression in rat limbic cortex, this improves neurogenic function, which is a pathophysiological base of anxiety and depression [77-79].

One of the EOs used in aromatherapy for the treatment of psychiatric disorders is lavender (*Lavandula angustifolia* Miller or *Lavandula officinalis* Chaix). Its composition is high in linalool and linalyl acetate content, precursors of THL. Investigations of possible mechanisms for lavender EO and its constituents have demonstrated a high affinity (dose-dependent) to the NMDA glutamate receptor with an IC_{50} value of 0.04 µl/mL (lavender oil) [80]. Studies have shown that linalool inhibits NMDA receptors, influencing (morphine-induced and conditioned) locale preference, acquisition, and reintegration; predisposing factors in opioid tolerance and dependence [81].

Molecular docking studies with amitriptyline (a tricyclic drug for depression and anxiety) and NMDAr at GluN1/

GluN2B, have revealed an energetically stable position, influenced by the tricyclic pose, and being stabilized in a V shape (two centralized aromatic rings at the hydrophobic part of the molecule situated at 120º from each other). Molecular dynamics simulations reveal that the hydrogen atom of the amine amitriptyline interacts with the N612 asparagine oxygen of one of the GluN2B subunits at a distance of 1.7Å. Stabilization within the channel is favored by hydrophobic interactions with uncharged aliphatic amino acids in the M3 helices of all NMDAr subunits. The receptor amino acids responsible for the drug interaction at position V are GluN1 (V642), GluN1 (T646), GluN2B (L640), and GluN2B (T644) [82].

NMDAr antagonists are of great interest for treatment of psychiatric conditions. Experimental data have shown that these receptors containing GluN2B subunits are responsible for about 80% of depolarizing currents in cortical neurons in primary cultures, which makes computational studies in determining and understanding drug coupling with these subunits important [83-85].

1.1.5. D1 and D2 Dopaminergic Receptors (PDB: 7CKX; 6CM4)

In animal models, mechanisms involving the mesolimbic dopaminergic system are related to depression and anxiety-like behaviors [86, 87]. The pathway is signaled by G protein-coupled metabotropic dopamine receptor activation (GPCR), which is subdivided into two main classes: D1-like (D1 and D5 - Gs ↑ cAMP), and D2-like (D2, D3, and D4 Gi - ↓ cAMP) [88].

The D1 receptor, when activated, mediates behavior related to anxiety, depression, reward, and memory, in addition to triggering immune and neuroinflammatory responses. Its mechanism can be stimulatory or inhibitory, depending on the region activated. In amygdala cells, inhibition of GABAergic interneurons (through D1 receptors), interspersed in para-capsular groups, increases angiogenesis and fear, whereas in basolateral regions, in projection neurons, the effect is excitatory [89]. As presynaptic D2 receptors, they are mostly observed in presynaptic prefrontal cortex terminals, and their activation exerts a GABAergic suppressive effect on parvalbumin-positive interneuron groups in amygdala cells. Local administration of D1 or D2 receptor agonists at the amygdala induces anxiety and depression-like effects, and anxiolytic and antidepressant effects as well [90].

Although GPCRs receptors are traditionally present in the form of monomers, it is currently known that there is a possibility of homo/hetero-tetrameric structure unions. Heteromeric dopamine receptor complexes such as D1-D2 present a discrete distribution, and their neurophysiological functions may differ from their individual constituent receptors [88]. Studies have shown that depressed and anxious patients tend to present a higher concentration of BDNF in the nucleus accumbens; resulting from increased expression of D1-D2. Importantly, BDNF/TrkB signaling presents differing effects (depending on the brain region) on the pathophysiology of anxiety and depression. In mesolimbic regions, its function can be depressant, whereas in regions such as the hippocampus (HC) and prefrontal cortex (PFC) its effect is antidepressant [91, 92].

Researchers have examined the effects of D1-D2 receptor activation and inhibition in rats, using the compounds SKF 83959 and TAT-D1 (respectively) in models of anxiety and depression induced by chronic disease and/or stress. Treatment with the D1-D2 stimulator, SKF 83959 increased immobility latency time in the forced swimming test, consumption of condensed milk, and reduced the time spent in the open arms in the EPM. These parameters predispose to depression and anxiety behavior (observed *in vivo*) and were reversed and/or minimized by treatment with TAT-D1 [87]. Experimental results revealed a higher density of D1-D2 heteromer complexes in the caudate nucleus of non-human primates and the striatum of female rodents. The results may imply greater susceptibility to phenotypic anxiety and depression in females. Modulation of the D1-D2 dopamine complex may also present a potential target for the treatment of anxiety and depression [93].

1.1.6. α_{2A} and α_{2B} Receptors (PDB: 6KUY and Swiss Model)

The central noradrenergic (NA) system has been implicated in many neuropsychiatric disorders. In α2-autoreceptors that are located in the presynaptic membranes of adrenergic neurons we see regulation of noradrenaline and adrenaline release [94]. On the other hand, α2-heteroreceptors are located in the presynaptic membranes of dopaminergic, serotonergic, and glutamatergic neurons. In postsynaptic neurons, α2 activation modulates neuronal excitability *via* regulation of ion channels, including direct modulation of potassium rectifier channels and indirect modulation of channels activated by hyperpolarization [95].

Coupled to a regulatory G protein, α-adrenoceptors are classified into α_{1A}, α_{1B}, α_{1D}, α_{2A}, α_{2B}, and α_{2C} subtypes. The α2 subtype receptors are linked to a Gi/o-type G protein which when activated, regulates intracellular adenylyl cyclase activity, negatively modulating the opening of Ca^{2+} channels, and Na^+/H^+-ATPase pumps, and positively modulating K^+ channel opening, limiting the release of catecholamines [96]. Studies in mice lacking the α2 subtypes clearly show that the functions these receptors play after being activated do not depend on a single subtype. However, most physiological and pharmacological functions of α2-adrenoreceptors are attributed to the α_{2A} subtype [94], and α2 adrenergic receptor agonists (such as clonidine) which present potent anesthetic, sedative, and antinociceptive properties [97].

The α_{2A} subtype is responsible for most "classic" functions mediated by α2-adrenoceptors, such as hypotension, sedation, hypothermia, anesthetic, and analgesic effects. Previous pharmacological studies had suggested that only the α_{2A} receptor mediates analgesic and anxiolytic effects [98], but evidence points to a contribution from α_{2B} subtype. The α_{2B} subtype shares several features observed in other G protein-$G_i/_o$-coupled receptor complexes, but the coupling specificity mechanism is not evident, though it is known that α2B-adrenoceptors performance is mediated by nitric oxide.

The binding properties of the α_{2A} and α_{2B} subtypes have been evaluated in *in silico* studies. Subtype A presents greater selectivity against ligands with bulky phenyl group substituents. Subtype B, on the other hand, seems to be less sensitive to the size of the ligand, since it has a larger opening. There is another structural feature that may also play a role in subtype selectivity. In the binding center, at position 5.31, subtype A presents glutamic acid, while subtype B presents lysine. Though the size of their side chains has no influence on the interaction with the ligands, their electrostatic properties, are very different, and this difference may have some effect on their ligand selectivity, as well as their hydrophobicity [99].

At the α_{2A} receptor binding site, substances that present imidazole rings bind at positions between helices 3 and 6 of the hydrophobic pocket, which present the residues Phe 6.52, Phe 6.53, and Val 3.33. The interaction also forms a bond between hydrogen and a Tyr 6.55 residue, as well as an electrostatic interaction with the Asp 3.32 side chain [100]. Molecular modeling studies have shown that aminophylline analogs interact with α2-adrenergic receptors. Mono and ortho-phenyl substituted compounds represent potentially new α_{2A} and α_{2B} agonists, being free of the side effects observed in agonists currently available on the market [101].

1.1.7. Noradrenaline and Serotonin Reuptake Carriers (Swiss Model, PDB: 68P_701)

Serotonin and norepinephrine (NA) are neurotransmitters that can be recaptured by serotonin (SERT) and norepinephrine (NET) transporters through a mechanism dependent on extracellular Na^+ and Cl^- (the driving transport force). The role of these transporters is to regulate neurotransmitter concentration (and consequent triggering responses) in the synaptic cleft. 5-HT as previously reported is a substance of great importance in the neurochemistry of depression and anxiety, and antidepressant drugs such as SSRIs induce increased proliferation of hippocampal cells and increased expression of neuroplasticity-related proteins such as BDNF [102, 103].

NA is produced from noradrenergic neurons that emerge from cell bodies at the *locus coeruleus* and project to different regions of the CNS (frontal cortex, amygdala, hippocampus, and hypothalamus) and spinal cord. At the level of the limbic system, the pathway is involved in emotion (appetite, pain response, pleasure, sexual satisfaction, and aggression) and cognition, which are compromised in patients with anxiety or depression. Generally speaking, the emotional and somatic centers of the brain receive projections from the NE and 5-HT pathways [104].

A number of SERT and NET inhibitors are used in anxiety and depression disorder pharmacotherapy, and as psycho-stimulants (drugs of abuse). Despite being commonly used and studied, their molecular binding mechanisms, inhibition, and selectivity are still not fully elucidated. Mutational analyses suggest binding in a region called pocket S1 and S2 (based on LeuT crystallographic structures) in an occluded substrate conformation that faces outside of the cell. Drugs such as (S)-citalopram present a potential binding conformation at the bottom of the S1 pocket, occupying all three pocket sub-sites [105]. Molecular docking studies suggest a basis for 5-HT transporter selectivity; being the presence of amino acid Asp-98 with an excess of negative charge located at Site A of unit S1. The bond formed is directly ionic with the 5-HT (and SERT inhibitor's) positive amino groups [106, 107].

Studies with linalool and β-pyrene (through *in vivo* models of anxiety and depression) have determined that these responses are due in part to activation of the serotoninergic pathway and modulation of adrenergic function through stimulation of noradrenergic receptors [27]. Yet, there currently is no conclusive data in the literature on the activity of these compounds as 5-HT and NA reuptake inhibitors. They therefore, remain important as targets in the pathophysiology of anxiety and depression, and of paramount importance in THL investigations.

2. MATERIAL AND METHODS

2.1. Chemicals and Drugs

Tetrahydrolinalool was purchased from Sigma-Aldrich® Chemical (St. Louis, MO, USA), diazepam (Cristália, Itapira, SP, BR) and imipramine (EMS, SP, BR). All substances were diluted in saline and orally administered (p.o.) at a total volume of 0.1 mL/10 g. THL was emulsified with Tween 80 (0.5%) in 0.9% saline.

2.2. Animals

Swiss mice (*Mus musculus*), albino adult male and female, weighing between 25-35 g, approximately 3 months old, kept under temperature control (21 ± 1 °C), with free access to water and feed, and a 12-hour light/dark cycle was used for all experimental protocols. The animals came from the Animal Production Unit (APU) of the Pharmaceuticals and Medicines Research Institute (IPeFarM) of the Federal University of Paraíba. All experimental procedures were previously approved by the CEUA - Ethics Committee on the Use of Animals at UFPB, under the certificate No. 6479250620.

2.3. *In vivo* Methodologies

2.3.1. Estimate of Acute Toxicity of Tetrahydrolinalool and Screening Pharmacological

The acute toxicity tests were performed according to the OECD "Guidelines for testing of chemicals" n. 423/2001 with modifications. Swiss mice, three females per group, including the control, were subjected to single doses of 300 and 2000 mg kg^{-1} of THL. The control group was treated with 0.9% saline + 0.5% Tween orally. After administration, observations were made at 30 min, 1 h, 2 h, 4 h, and then once daily until the fourteenth day. On the fourteenth day, the animals will be weighed, and then euthanized with an injection of anesthetics, as recommended by the Euthanasia Practice Guidelines of the National Council for Animal Experimentation Control (CONCEA), with the organs (liver, spleen, heart, kidneys, and brain) removed, weighed and evaluated macroscopically. THL doses from subsequent pharmacological trials were established from the estimated LD_{50} value.

2.3.2. Elevated Plus Maze Test

Groups of mice were used (n = 6). Substances, vehicle (control group: Tween 80 - 0.5%), diazepam (1 mg kg^{-1}), or THL (37.5 - 600 mg kg^{-1}), administered 30 (diazepam) and 60 (THL) min before the start of the test. The standard group, treated with diazepam, was administered i.p, and for the others, the recommended route was oral. After 30 or 60 minutes, each animal was placed individually on the central platform of the device, facing an open arm. The observed behavioral parameters were the number of entries and length of stay (in seconds) in the open arms. An entry was counted only if the animal had four legs outside of the central line of the labyrinth. Each animal remained for a time of 5 min in the maze [108].

2.3.3. Open Field Test

The animals were submitted to the open field test, where spontaneous individual exploration and motor activity were evaluated. The control group was replaced by diazepam (2 mg kg^{-1}, i.p.). The open field constitutes a circular base of 30 cm in diameter, marked in an internal and external circumference and an acrylic dome measuring 30 cm in height x 30 cm in diameter. For 5 minutes, the following parameters were recorded: frequency of locomotion or number of crossings in the quadrants, number of surveys or rearing (lifting the animal on its hind legs), self-cleaning or grooming number [109].

2.3.4. Rotarod

To assess the possible non-specific muscle relaxation effects of the compounds, the animals were subjected to the route rod test. The animals were exposed to the apparatus 24 h before the test, and the animals were managed to stay on the rotating bar (at 10 rpm) for 1 minute and were selected for testing. As a control group, diazepam was used in the dose of (4 mg kg^{-1}i.p). At 60 and 120, minutes, each animal was placed on a rotarod apparatus, and the time (in seconds) completed on the bar was measured for up to 180 sec [110, 111].

2.3.5. Forced Swimming Test

The swimming forced tester, proposed by Porsolt, has been used as a predictive model of antidepressant-like activity. Groups of mice were used (n = 6). Substances, vehicle (control group: Tween 80 - 0.5% p.o.), imipramine (30 mg kg^{-1} i.p.), or THL (37.5 - 600 mg kg^{-1}p.o), administered 30 (imipramine) or 60 (THL) min before the start of the test. Mice were placed, one at a time, in the cylinder, and immobility time, in seconds, was recorded for 5 min. Immobility was defined as the absence of active, escape-oriented behaviors, such as swimming, jumping, rearing, sniffing, or diving [112]. Any mouse appearing to have difficulty keeping its head above water was removed from the cylinder and excluded from the analysis.

2.4. *In silico* Methodologies

2.4.1. Protein Sequence Alignment

The protein sequences that did not contain 3D structures from the Protein Data Bank (PDB https:www.rcsb.org/pdb/ho me.do) were obtained from the GenBank database (https://ww w.ncbi.nlm.nih.gov/) [113].The proteins: Noradrenaline transporter (NP_001165975.1) and α_{2B} receptor (NP_000673.2). Global alignment, using the web tool Clustal Omega (https://www.ebi.ac.uk/Tools/msa/clustalo/) is performed with the sequence of the protein with a known three-dimensional structure, which then aligns all protein sequences entered by the user. Alignment facilitates investigation of the active site and determination of similarity and shared identity between proteins.

2.4.2. Homology Modeling

Target sequences were obtained as amino acid sequences in the FASTA format and were imported from the Swiss Model website (https://swissmodel.expasy.org/) [114]. For each mold identified, the quality was predicted from alignment features such as ProMod3, QMEAN, and GMQE. The stereo-chemical quality of the models was assessed on the PSVS web server (protein structure validation software suite) (http://psvs-1_5-dev.nesg.org/), using Procheck [115]. Procheck generates a Ramachandran plot, which determines the permitted and disallowed regions of the main amino acid chain [116].

2.4.3. Molecular Docking

Molecular docking was used to investigate the mechanism of action of tetrahydrolinalool (THL) which yields its anxiolytic and antidepressant effects as a result of binding affinities between the compounds and the enzymes selected. For this, screening was performed with several of the proteins involved. The 3D structures of the enzymes were obtained from the Protein Data Bank (PDB) (https://www.rcsb.org/pdb /home/home.do) [113]. The proteins selected and detailed information about them can be seen in Table **1** (Supplementary Material **S1**). Initially, all water molecules were removed from the crystal structure, and the root mean square deviation (RMSD) was calculated from the poses; indicating the degree of reliability of the fit. RMSD provides a connection-mode close to the experimental structure and is considered successful if the value is less than 2.0 Å.

2.4.4. Molegro Virtual Docker (MVD) 6.0

The Molegro Virtual Docker v.6.0.1 (MVD) software was used with the predefined parameters in the same software. The complexed ligand was used to define the active site, and the compounds were then imported to analyze the

Table 1. Receptors selected in the study.

S. No.	Protein	PDB ID/Homology	Binder	Resolution
1.	Serotonin transporter	5I73	Escitalopram	3.24 Å
2.	Noradrenaline transporter	Homology model	Atomoxetine	-
3.	Neuronal nitric oxide synthase	5UO1	L-name	1.90 Å
4.	Gyanylate cyclase soluble	3UVJ	Methylene blue	2.08 Å
5.	NMDA receptor	4PE5	Ketamine	3.96 Å
6.	TNF-α inhibitor	2AZ5	PDB binder	2.10 Å
7.	IL-6 inhibitor	1ALU	Methotrexate	1.90 Å
8.	D1 receptor	7CKX	SCH cloridrate	3.54 Å
9.	D2 receptor	6CM4	Risperidone	2.87 Å
10.	α2A receptor	6KUY	Prazosin	3.20 Å
11.	α2B receptor	Homology model	Prazosin	-
12.	5-HT1$_B$ receptor	4IAQ	Dihydroergot	2.80 Å
13.	5-HT1$_A$ receptor	7E2X	Buspirone	3.0
14.	5-HT2$_A$ receptor	6A93	Risperidone	3.00 Å

stability of the system through interactions identified at the active site of the enzyme, taking as reference energy the MolDock value. The MolDock SE (Simplex Evolution) algorithm was used with the following parameters: a total of 10 runs, with a maximum of 1500 iterations, using a population of 50 individuals, with 2000 minimization steps for each flexible residue, and 2000 global minimization steps per run. The MolDock Score (GRID) scoring function was used to calculate the snap energy values. A GRID was set at 0.3 Å, and the search sphere was set at a 15 Å radius. For analysis of the ligand energy, internal electrostatic interactions, internal hydrogen bonds, and sp^2-sp^2 torsions were evaluated.

2.4.5. Pharmacokinetic Predictions

Pharmacokinetic predictions (available for free) were obtained from electronic platforms: SwissADME and pkCSM [117, 118]. For predictive model interfacing, the structures were generated by the Chemdraw Ultra 12.0 software into the smile format.

2.5. Statistical Analysis

The results obtained during *in vivo* experiments were analyzed using one-way analysis of variance (ANOVA) software, followed by the Tukey test to compare the means. Data were expressed as mean ± S.E.M. (standard error of the mean), and a confidence level of 5% was adopted.

3. RESULTS AND DISCUSSION

3.1. Pharmacological Behavioral Screening and LD_{50} Estimate

The results demonstrate that under the conditions evaluated, no deaths, and no morphological, physiological, and/or behavioral changes were identified during the observation period in the animals treated with a dose of 300 or with a dose of 2000 mg kg^{-1}(Supplementary material - **S2**).

The compound THL per v.o. presented low toxicity, with no evidence of changes in organic functions, eyes, hair, skin, (a priori in class 5), estimating that the LD_{50} is found in values above 2000 mg kg^{-1}, and less than 5000 mg kg^{-1}. In the literature, single-dose acute toxicity studies with THL determined that the calculated LD_{50} was 6.233 *g* kg^{-1} (± 0.498 g/kg) [119]. Thus, it was possible to establish a set of safe doses for realization of subsequent *in vivo* pharmacological tests [120].

3.2. Assessment of THL's Anxiolytic-like Potential

3.2.1. Elevated Plus Maze Test (EPM)

The results presented in (Fig. **2A**) revealed that animals pre-treated with THL at doses of 37.5 (9.2 ± 0.9) and 75 (7.0 ± 1.0) mg kg^{-1} significantly increased the number of entries into the open arms of the maze (NEOA), with percentages of 72.7 and 64.3% respectively, compared to the control group (2.5 ± 0.9). In the group treated with DZP 1 mg kg^{-1} (10 ± 1.3), an increase of 75% was observed compared to the control group.

Regarding the length of stay in the open arms (TPOA) (Fig. **2B**), THL significantly increased the parameter in question at doses of 37.5 (127.5 ± 8.4), 75 (108.5 ± 15.9), and 150 (115.2 ± 9.9) mg kg^{-1}, respectively by 80.3, 76.8, and 78.2%, as compared to the control group (25.2 ± 10.2). A similar effect was obtained with DZP pretreatment at 1 mg kg^{-1} (90 ± 8.7), which increased TPOA by 72.2% as compared to the control.

Due to the lack of psychopharmacological studies using THL, the scientific reporting on its precursor linalool was

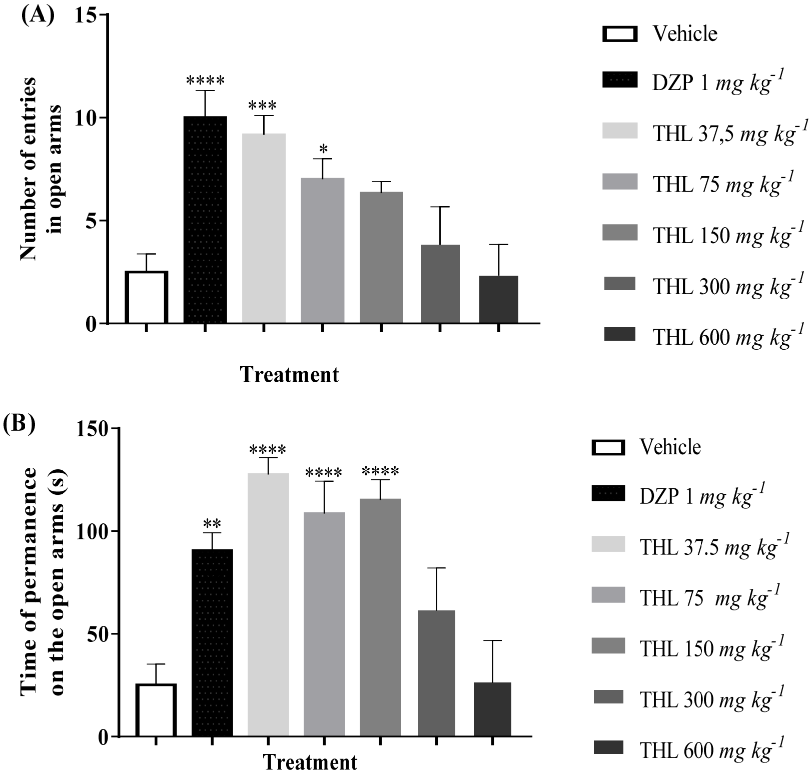


Fig. (2). Effect of THL at doses of (37.5 - 600 mg kg^{-1}p.o) and DZP (1 mg kg^{-1}i.p), on the number of entries (**A**) (F value = 11.84) and time permanence (**B**) (F value = 16.8) of the animals in the open arms of the elevated plus maze. The column represents the average ± S.E.M. (n = 6). Statistical analysis: ANOVA unidirectional followed by the Tukey test, **p*<0.05, ***p*<0.01, *** *p* <0.001, *****p*<0.0001: (THL) = *vs*. control and (DZP) = *vs*. control.

taken as a basis. Research has shown that its intraperitoneal or oral systemic administration induces anxiolytic effects in classical models that assess anxiolytic-like activity *in vivo* [5, 27, 121, 122]. In the EPM model, the increase in NEOA and TPOA by rodents is indicative of anxiolytic drugs, considering that the animals feel safe when exploring. Studies have evaluated the effects of lavender essential oil (LEO) from *C. osmophloeum* ct. (at 250 and 500 mg kg^{-1}), and its main compound, (in enantiomer forms) S - (+) - linalool (500 mg kg^{-1}). Significant results were observed (for all forms) being as effective as trazodone hydrochloride, the reference drug used as a standard [13].

3.2.2. Open Field Test (OFT)

In a new environment, the animal's natural tendency is to explore it, despite the stress and conflict caused by exposure to the environment. The open-field test aimed to evaluate spontaneous locomotion, as well as the grooming (self-cleaning) and rearing (vertical exploration) behavior of animals treated with THL at doses of 37.5 - 600 mg kg^{-1}. The results shown in Fig. (**3A**) demonstrate that the monoterpene at doses of 37.5 (60.5 ± 3.8), 75 (55.2 ± 1.7), and 150 (78.8 ± 6.0) presented no significant change in the number of crossings in the Open Field, showing that, in this dose range, THL does not induce sedation in the animal. On the other hand, the groups treated with THL at doses of 300 (48.7 ± 3.1) and 600 mg kg^{-1} (48 ± 2.1) as well as with DZP at *2* mg kg^{-1} (43.7 ± 5.1) displayed significantly impaired locomotor activity, and a reduced number of crossings (by 30.6, 31.6, and 37.8 %, respectively), when compared to the control group (70.2 ± 1.9) treated with the vehicle.

Fig. (**3B**) presents the results obtained in relation to the number of groomings (self-cleaning). No significant changes were observed in the groups treated with THL or diazepam when compared to the control group (vehicle). On the other hand, in Fig. (**3C**), we observe that there was a significant increase in the number of rearings (vertical exploration) for the groups treated with THL at doses of 37.5 (22.2 ± 1.3) and 75 (22.2 ± 1.3) mg $kg^{-1}$by 42% when compared to the control group (12.8 ± 1.4). The group treated with DZP at 2 mg kg^{-1} (6.3 ± 0.8) presented a reduction in the number of rearings of 50.7 %. This was possibly due to the sedation caused by the Diazepam dosage. The other doses of THL did not show significant differences in relation to these parameters when compared to the control group (vehicle).

3.2.3. Rotarod Test (RRT)

The results obtained demonstrated that the latency to fall of the groups treated with the monoterpene, tested after 60 and 120 minutes, at the doses of 37.5 - 600 mg kg^{-1} was not significantly different from those in the vehicle-treated group. The animals in the standard group, DZP (4 mg kg^{-1}) (62.2 ± 13.2) did not remain on the rotating bar during the evaluation in the first hour (Fig. **4**).

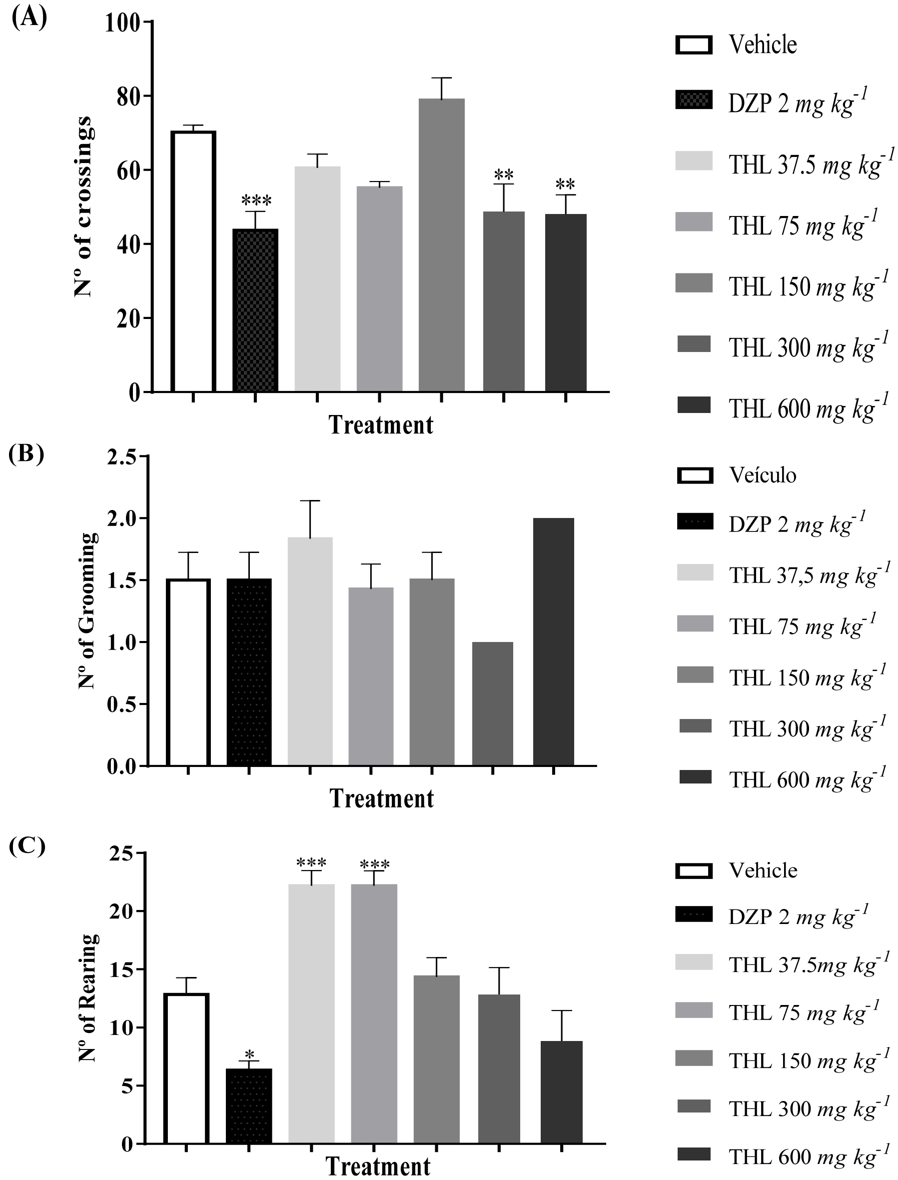


Fig. (3). Effect of THL at doses of (37.5 - 600 mg kg^{-1}p.o) and DZP (2 mg kg^{-1}i.p), on the number of crossings (**A**) (F value = 11.78), number of grooming (**B**) (F value = 2,42), and number of rearing (**C**) (F value = 23.42) on open field test. Statistical analysis: ANOVA unidirectional followed by the Tukey test, **p*<0.05, ***p*<0.01, ****p* <0.001, *****p*<0.001: (THL) = *vs*. control and (DZP) = *vs*. control.

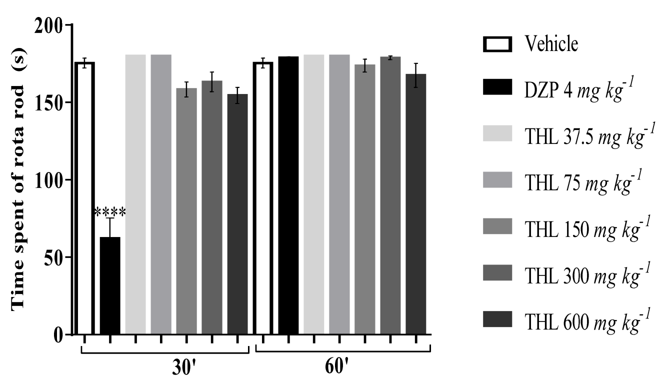


Fig. (4). Effect of THL at doses of (37.5 - 600 mg kg^{-1}p.o) and DZP (4 mg kg^{-1}i.p), the time of stay (s) of the animals on the rotarod. The column represents the average ± S.E.M. (n = 8). Statistical analysis: ANOVA unidirectional (F value = 35.06) followed by the Tukey test, **p*<0.05, ***p*<0.01, ****p*<0.001, *****p*<0.001: (THL) = *vs*. control and (DZP) = *vs*. control.

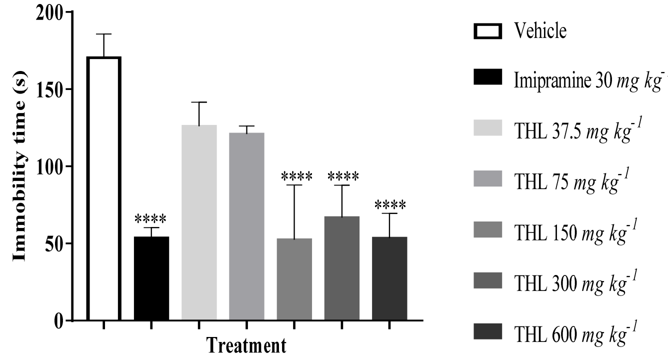


Fig. (5). Effect of THL at doses of (37.5 - 600 mg kg^{-1}p.o) and Imipramine (IP) (30 mg kg^{-1}i.p), the time of immobility (s) of the animals on forced swim test. The column represents the average ± S.E.M. (n = 8). Statistical analysis: ANOVA unidirectional followed (F value = 17.74) by the Tukey test, **p*<0.05, ***p*<0.01, ****p* <0.001, *****p*<0.001: (THL) = *vs*. control and (IP) = *vs*. control.

To minimize error in behavioral interpretations, the model used to assess the influence of substances on an animal's locomotor activity and motor coordination is of critical importance [123-125]. Studies with *Cinnamomum osmophloeum* ct. EO, presenting linalool as a major compound, revealed no significant differences in latency to fall in the RRT between groups treated orally with leaf oil EO at a dose of 100 mg kg^{-1}. However higher doses (200 and 400 mg kg^{-1}) increased the time spent on the swivel bar [126]. Further, derivatives such as linalool oxide did not affect motor coordination when administered intraperitoneally at doses of 50, 100, and 150 mg kg^{-1} [127]. The results obtained with THL are in agreement with its precursor, linalool, suggesting that unlike benzodiazepines, whose use may be restricted, the anxiolytic activities of THL and its precursor linalool, do not negatively affect motor coordination, [128, 129].

3.3. Assessing the Antidepressant-like Potential of THL

The forced swim test (FST) is a behavioral model with predictive validity, widely used to assess the antidepressant activity of new compounds [112]. The parameter measured in the test is immobility time (IT), characterized by the animal's lack of movement, except for what is necessary to keep its head above the water level [130]. Acute exposure to THL in mice treated with doses of 150 (52.33 ± 14.5), 300 (66.67 ± 8.6), and 600 mg kg^{-1} (53.3 ± 6.6), significantly reduced the respective immobility times by 69.3, 60.9 and 68.7%. The results were comparable to those obtained by the group treated with imipramine 30 mg kg^{-1} (53.5 ± 6.8), which decreased the IT by 68.6. All groups were compared with the control (170.3 ± 15.4), and treated with the vehicle (Fig. **5**).

Due to their antidepressant properties, in the Amazon region, aromatic, linalool-rich EO-producing plants are extensively used in traditional medicine. *Aniba rosaeodora* Ducke (rosewood, rosewood), *Aniba parviflora* (Meissn.) Mez (macacaporanga), and *Aeollanthus suaveolens* Mart. ex Spreng (catinga-de-mulata) are included among the species [10]. Studies with these EO and with a linalool standard (30 mg kg^{-1}) demonstrated significant effects in forced swimming models at the highest doses (35, 85, and 75 mg kg^{-1}); significantly decreasing the immobility time of the animals, when compared to the controls. Linalool presents an effect in models that simulate anhedonia, such as the splash test that assesses the desire for self-cleaning [10, 131]. Anhedonia is characterized by a loss of interest in the development of daily activities and is considered a central symptom in the diagnosis of depressive disorders [132-134].

Other data also corroborate the potential of linalool and linalyl acetate, major compounds in a series of EOs, such as the lavender (*Lavandula angustifolia* Mill), used in therapy to control depressive disorders and anxiety. Lavender EO is commercially available under the trade name Silexan® in capsules [135]. The drug is presented in doses of 80 and 160 mg, and its functionality has been scientifically proven in cases of Generalized Anxiety Disorder (GAD) and Major Depressive Disorder (MDD) comorbidity (ICD-10 F41.2) [136]. Its mechanisms of action involve modulation of voltage-dependent calcium channels with influences on CREB-mediated neuroplasticity, and activation of intracellular signaling kinases such as PKA and MAPK [135].

The psychopharmacological studies discussed in this brief review with linalool demonstrate its anxiolytic and antidepressant potential in modulating various signal pathways. The possibility of evaluating the anxiolytic and antidepressant-like activities of THL demonstrated satisfactory effects in classical models *in vivo*. The later stages of this study employed computational tests as a fundamental basis for determining and investigating potential molecular response mechanisms through the pharmacological activity of this derivative.

3.4. *In silico* Tests

3.4.1. Protein Sequence Alignment

Protein sequence alignment is a tool that helps verify the similarity and identity of the same protein from different species, or different proteins from the same species. With this technique, it is possible to analyze conserved regions and identify common residues in the active site. Further, it is possible to point out differences and structural similarities that may contribute in later stages of drug development. Amino acids shared between target protein sequences and

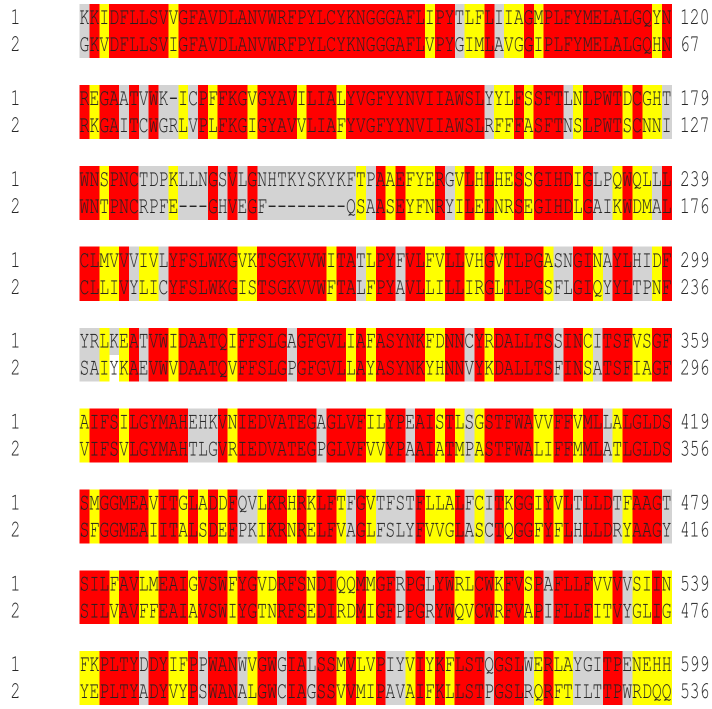


Fig. (6). Sequence alignment of the noradrenaline transporter target from *Homo sapiens* (**1**) and the Dopamine 1 template from *Drosophila melanogaster* (**2**). Gray regions correspond to non-similar and non-identical amino acids. Red regions correspond only to identical amino acids. The yellow regions are similar amino acids. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

template protein sequences were investigated. The results revealed that the Homo sapiens noradrenaline transporter achieves 59.27% identity with Dopamine 1 from Drosophila melanogaster (Fig. **6**), and the H. sapiens α_{2B} receptor achieves 97.76% identity with the Bos taurus α_{2B} receptor (Fig. **7**). The alignment results revealed that the target protein sequences present a high degree of identity and similarity, which enabled the construction of reliable homology models for these proteins.

3.4.2. Homology Modeling

The noradrenaline transporter and α_{2B} receptor model was generated using homology modeling methods. The reliability of the models was assessed using the Ramachandran plot, which represents all possible combinations of dihedral angles Ψ (psi) *versus* φ (phi) for each amino acid in a protein, except glycine, which has no side chains. The models are considered reliable when more than 90% of the amino acids are present in the allowed and/or favored regions (colored regions of the graph). Blank regions represent outliers. The noradrenaline transporter model presented 92.2% of the amino acids in the favored regions, and 6.3% in allowed regions (Fig. **8A**). The α_{2B} receptor model presented 90.3% of the amino acids in the favored regions, and 8% in allowed regions (Fig. **8B**). Given the results, the homology models were considered reliable.

3.4.3. Molecular Docking

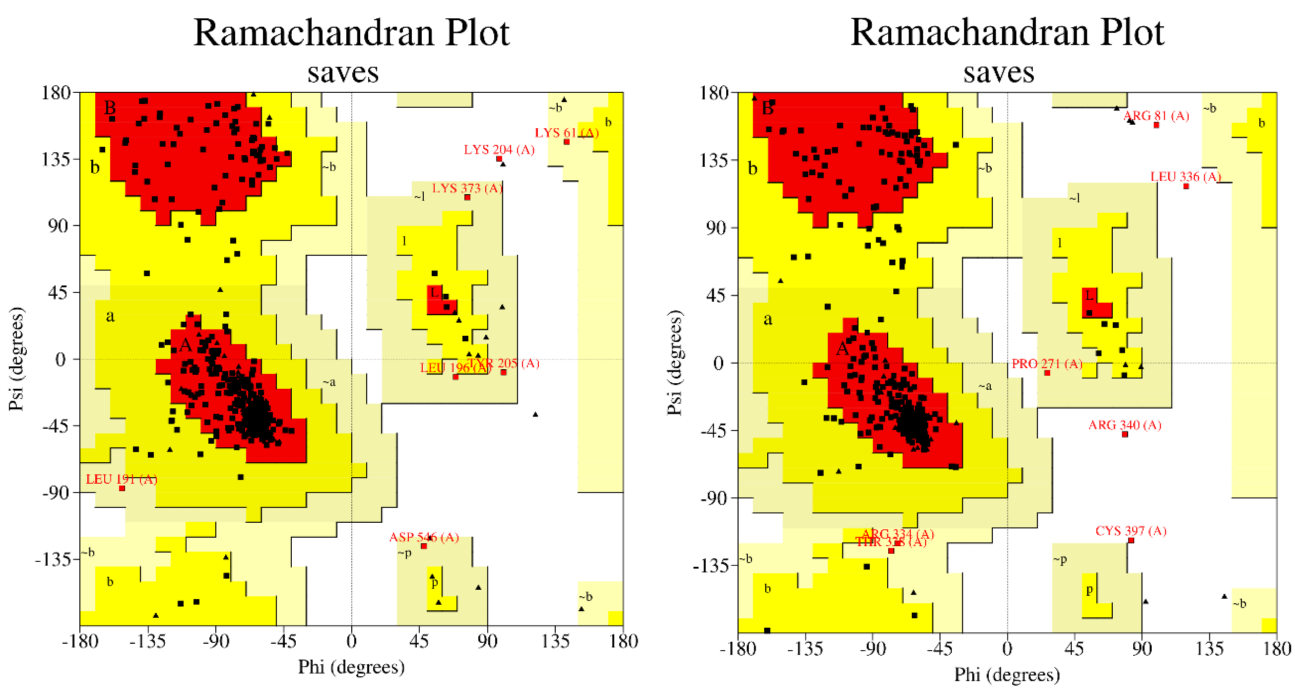
The compound tetrahydrolinalool (THL) was submitted to screening by molecular docking in 14 proteins. The docking results were generated using two scoring functions, MolDock score and Rerank score. For most scoring functions, the more negative values indicate better predictions. A protein-compound binding energy value of greater than or close to the standard drug in at least one scoring function is considered active.

The docking results generated by the five scoring functions are validated by redocking the crystallographic ligand with all investigated proteins. The root mean square deviations (RMSDs) of the obtained fit poses are then calculated against the crystal structure. RMSD values less than 2 Å indicate an optimal degree of screening reliability. Information about the starting structures and the results of the redocking validation is shown in Table **S3**. During the redocking analysis, most of RMSD values were below 2.0 Å, that is, the generated poses correctly positioned the ligand in the active site. In general, for docking validation, the programs provided values considered satisfactory.

Computational techniques, when applied to the development and discovery of new drugs, allow focusing and optimizing time, work, and operating costs. These techniques have become the initial phase for directing both preclinical trials and our understanding of various molecular mechanisms and biological phenomena. Molecular docking is an approach that allows us to understand the processes of coupling and interactions between a molecule and a protein at the atomic level. The calculations generated allow predicting the pose, orientations, and the affinity of the ligand to the active sites of the target. Our studies were able to characterize the THL complex in important targets in CNS

```
1   ------------------------------------------------------------ 0
2   IYKDTEGYYTIGIGHLLTKSPSLNAAKSELDKAIGRNTNGVITKDEAEKLFNQDVDAAVR 120

1   ------------------------------------------------------------ 0
2   GILRNAKLKPVYDSLDAVRRAALINMVFQMGETGVAGFTNSLRMLQQKRWDEAAVNLAKS 180

1   -------------------MDHQDPYSVQATAAIAAAITFLILFTIFGNALVILAVLTSR 41
2   RWYNQTPNRAKRVITTFRTGTWDAYYSVQATAAIAAAITFLILFTIFGNALVILAVLTSR 240

1   SLRAPQNLFLVSLAAADILVATLIIPFSLANELLGYWYFRRTWCEVYLALDVLFCTSSIV 101
2   SLRAPQNLFLVSLAAADILVATLIIPFSLANELLGYWYFRRTWCEVYLALDVLFCTSSIV 300

1   HLCAISLDRYWAVSRALEYNSKRTPRRIKCIILTVWLIAAVISLPPLIYKGDQGPQPRGR 161
2   HLCAISLDRYWAVSRALEYNSKRTPRRIKCIILTVWLIAAVISLPPLIYKGDQGPQPRGR 360

1   PQCKLNQEAWYILASSIGSFFAPCLIMILVYLRIYLIAKRSNRRGPRAKGGPGQGESKQP 221
2   PQCKLNQEAWYILASSIGSFFAPCLIMILVYLRIYLIAKRSNRRGPRAKGGPGQG----- 415

1   RPDHGGALASAKLPALASVASAREVNGHSKSTGEKEEGETPEDTGTRALPPSWAALPNSG 281
2   ------------------------------------------------------------ 415

1   QGQKEGVCGASPEDEAEEEEEEEEEEEEECEPQAVPVSPASACSPPLQQPQGSRVLATLRG 341
2   ------------------------------------------------------------ 415

1   QVLLGRGVGAIGGQWWRRRAQLTREKRFTFVLAVVIGVFVLCWFPFFFSYSLGAICPKHC 401
2   ------------EQWWRRRAQLTREKRFTFVLAVVIGVFVLCWFPFFFSYSLGAICPKHC 463

1   KVPHGLFQFFFWIGYCNSSLNPVIYTIFNQDFRRAFRRILCRPWTQTAW            450
2   KVPHGLFQFFFWIGYCNSSLNPVIYTIFNQDFRRAFRRILCRPWTQTAW            512
```

Fig. (7). Sequence alignment of the *H. sapiens* α_{2B} receptor target (**1**), and the α_{2B} receptor template from *Bos taurus* (**2**). Gray regions correspond to non-similar and non-identical amino acids. Red regions correspond only to identical amino acids. The yellow regions are similar amino acids. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).



(**A**) (**B**)

Fig. (8). Ramachandran plot of the homology model generated for the noradrenaline transporter (**A**) and α_{2B} receptor (**B**). The colored regions represent the permitted and favored regions of the secondary structures, and the white regions represent prohibited regions. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

modulation, those which impact the pathophysiology of anxiety and depression [137].

According to the data obtained (Table **2**) from the 14 analyzed proteins, THL presented a significant interaction with the proteins nNOS, SGC, NMDA, IL-6, D1, and 5-HT_{1A}. We observed that the compound obtained values higher than or close to the Rerank score values when compared to the controls. THL was most potent in targeting

Table 2. Binding energy values analyzed for the 14 proteins selected in the study. The best results are highlighted in bold.

Protein	THL		Positive Control	
	Moldockscore	Rerankscore	Moldockscore	Rerankscore
Serotonin Transporter	-65.38	-55.64	-148.39	-114.03
Noradrenaline Transporter	-78.14	-65.02	-107.98	-85.73
nNOS	-42.80	-35.99	-46.21	-44.33
GC	-61.73	-51.35	-95.24	-63.94
NMDA	-80.16	-64.67	-80.69	-52.32
TNF-α	-73.94	-59.49	-146.69	-83.99
IL-6	-39.99	-23.25	-86.46	-38.23
D1	-55.05	-44.46	-63.43	16.58
D2	-83.32	-70.09	-153.86	-133.46
α_{2A} receptor	-57.29	-49.23	-129.01	-110.33
α_{2B} receptor	-53.77	-47.00	-123.18	-103.93
5-HT_{1B} receptor	-75.28	-59.86	-216.217	-175.03
5-HT_{1A} receptor	-72.41	-57.34	-105.95	-56.63
5-HT_{2A} receptor	-78.83	-66.49	-146.93	-124.58

NMDA and D1 proteins, with respective binding affinity values of -64.67 and -44.46 kcal/mol. Similar results can be observed in computational studies with the phenylpropanoids 4-allyl-2,6-dimethoxyphenol, methyl eugenol, and ortho eugenol, which demonstrate antagonistic potential in targets involved in depression, such as NMDAr and nNOs [138]. For the nNOS, GC, IL-6, and 5-HT_{1A} proteins, tetrahydrolinalool presented energy values close to that of the crystallographic ligand, indicating probable activity with these enzymes, but with a weaker interaction. The results obtained for THL suggest both its potential and possible mechanisms of action.

Interactions with proteins in which THL obtained a higher binding affinity results than the standard drug were observed in detail. Interactions and bonds formed between THL and the proteins NMDA and D1 were analyzed. THL was able to form three hydrophobic interactions with the amino acids Pro78, Ala107, Tyr109, Ile111, and Phe114 at the active site of the NMDA protein (Fig. **9**). Ketamine formed a hydrogen bond with Gln110, three hydrophobic interactions with the amino acids Tyr109, Ile111, and Phe114, and an unfavorable, steric-type interaction with the amino acid Ala107. In the D1 receptor, THL presented several hydrophobic interactions with the active site, including Val110, Ile104, Leu190, Trp285, Phe288, and Phe289 (Fig. **10**). The standard presented a hydrogen bond with Ser189, a van der Waals bond with Asp103, and hydrophobic interactions with the residues Val100, Ile104, Leu190, and Phe 288.

3.4.4. Pharmacokinetic Predictions

The pharmacokinetic mechanisms of absorption, distribution, metabolism and excretion (ADME) have been noted as principal causes of failure when investigating new drug candidate molecules [139]. Modern *in silico* modeling methodologies that assess pharmacokinetic properties are important during initial studies, optimizing the research process for new drugs, and identifying the principal ADME problems that may compromise later phases of preclinical research [140]. In this context, among the alternative computational tools, SwissADME and pkCSM offer access to a pool of robust and predictive models for physical-chemical, pharmacological, and medicinal chemistry properties [118, 141, 142].

To identify potential THL pharmacokinetic failures, predictions were performed on the aforementioned platforms. Initially, Lipinski's parameters were observed, whose rules determine the chance of the compound being potentially promising with oral availability. The criteria are: (a) molecular weight ≤ 500 da; (b) LogP ≤ 5 (or MLogP ≤ 4.15); (c) number of hydrogen bond acceptors ≤ 10; and (d) number of hydrogen bond donors ≤ 5. THL has a MLogP of 2.89 and adequate numbers (based on Lipinski's rule) of hydrogen bond donors and acceptors to suggest good oral availability. Regarding LogS (-2.55) the values were within the (-4 to -2) range, which indicates good solubility in water.

In the boiled egg's model, THL (Fig. **11**) presented a high probability of being absorbed in the gastrointestinal tract and of permeating the blood-brain barrier (BBB) (pkCSM: logBB = 0.629 > 0.3), which can be graphically visualized as points referring to the physical-chemical area of the brain (the yolk). It can also be seen that THL is not a substrate or inhibitor of P-glycoprotein (P-gp) or multi-drug resistance protein (MDR1) (Table **3**). This information is relevant due to the role of P-gp as an efflux pump, which prevents the substance from reaching therapeutic concentration in the human body or permeating the HE barrier [143, 144].

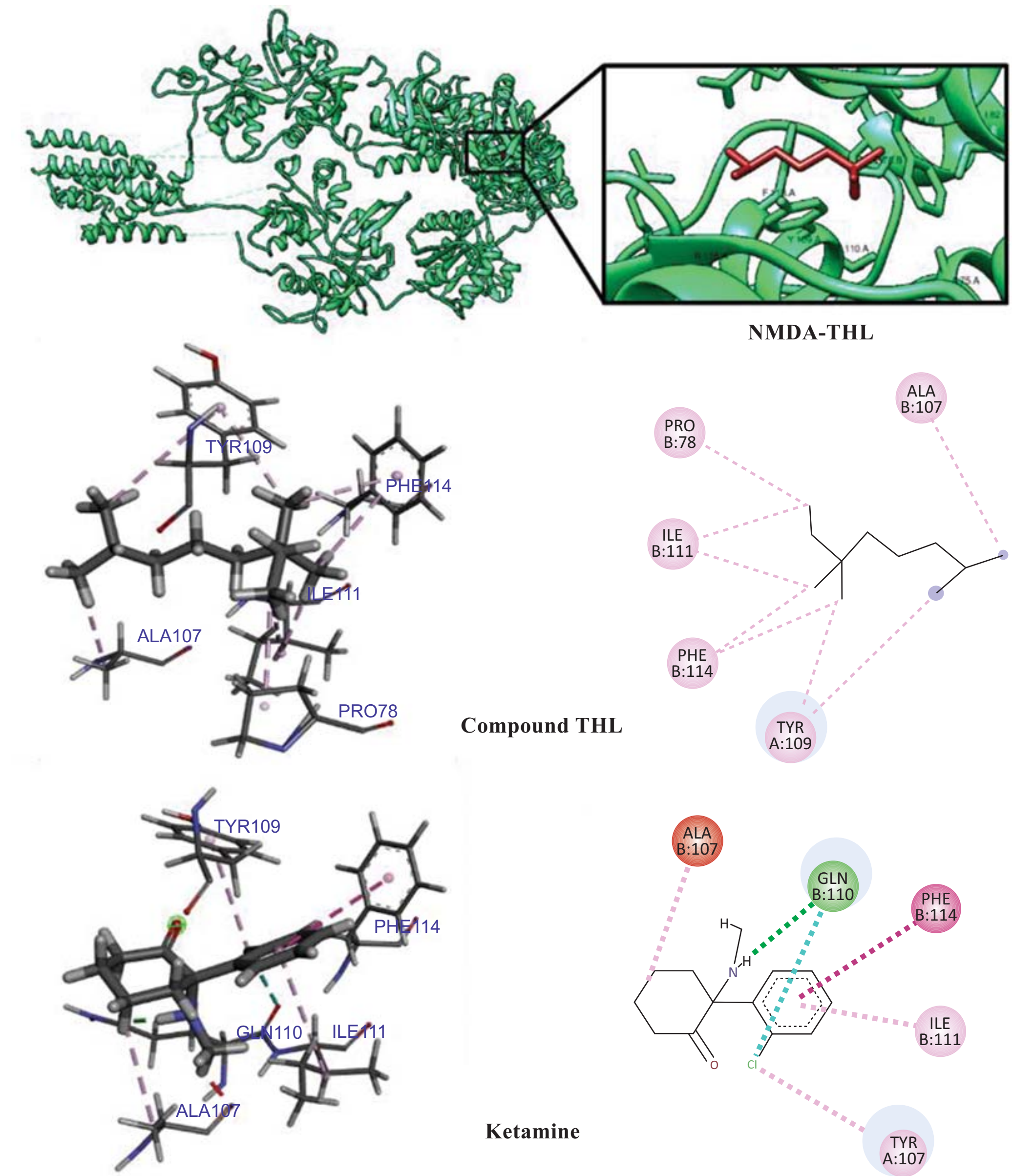


Fig. (9). 2D and 3D interactions between THL, Ketamine, and the protein NMDA. Hydrogen bonds are highlighted in green; hydrophobic interactions are highlighted in pink, and steric interactions are highlighted in red. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

Table 3. THL: Potential *in silico* inhibition of cytochrome P450 complex isoforms. Source: Adapted from SwissADME and pkCSM (2021).

(CYP)	THL
CYP2D6 (Substrate)	No
3A4 (Substrate)	No
1A2 (Inhibition)	No
2C19 (Inhibition)	No
2C9 (Inhibition)	No
2D6 (Inhibition)	No
3A4	No

Source: adapted from *SwissADME* and pkCSM (2021).

Analysis in pkCSM, (using *in silico* Caco-2 adenocarcinoma cell permeability values) obtained logarithm parameters for apparent permeability coefficient values of above 90% (log Papp, log cm s-1 > 0.9) indicating the high probability of THL absorption in the human small intestine. The data corroborate those obtained using the Swiss ADME platform (TPSA ≤ 140 Å) (Table **4**). The steady-state volume of distribution (VDss) is a theoretical volume that a global dose of a drug would need to be uniformly distributed (ensuring balanced plasma concentration), here THL presents low values, which might interfere with an adequate distribution profile. Clearance rate influences bioavailability and its determination is needed to establish steady-state dosage concentrations [145, 146].

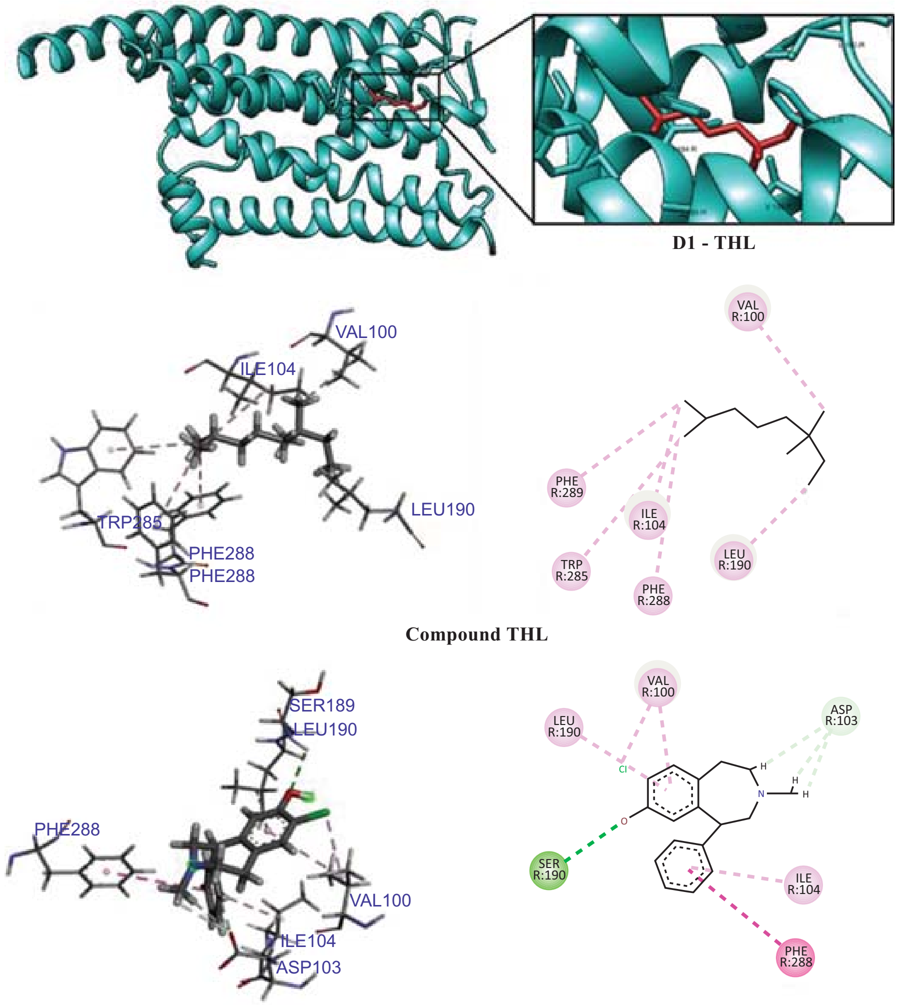


Fig. (10). 2D and 3D interactions between THL, SCH hydrochloride and the protein D1. Hydrogen bonds are highlighted in green; hydrophobic interactions are highlighted in pink, and steric interactions are highlighted in red. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

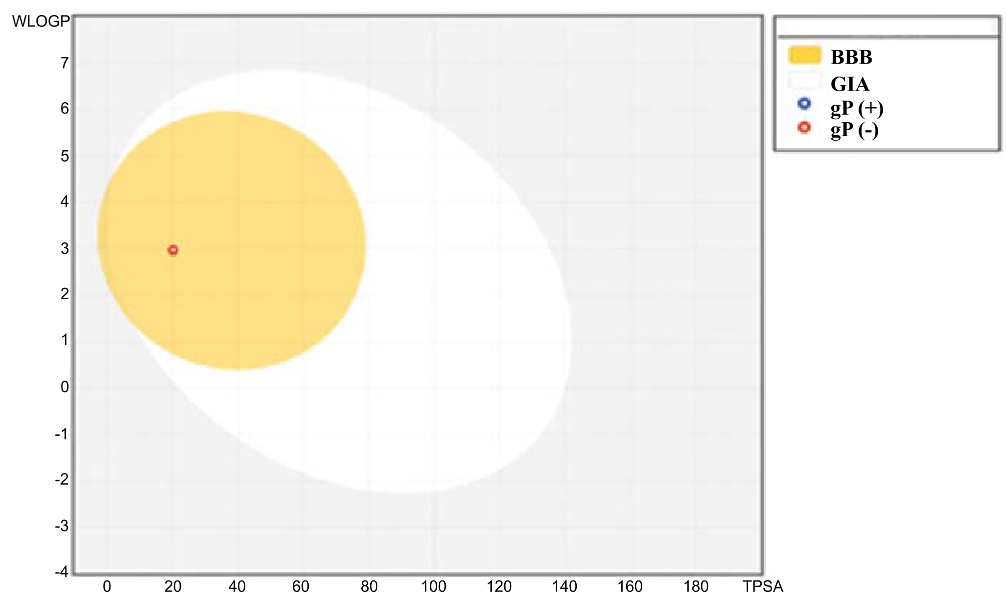


Fig. (11). Absorption profile of tetrahydrolinalool (THL) using the (Boiled-egg) *in silico* model *via* SwissADME. GIA: gastrointestinal absorption; gP (G+ or - protein substrate). Source: SwissADME (adapted) (2021). (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

Table 4. ***In silico*** **pharmacokinetic data as estimated using the SwissADME and pkCSM platforms.**

Compound	Log*P*	Log*S*[b]	TPSA (Å^2)[c]	P. Caco-2 (log Papp; log cm/s)[d]	Int. Ab. (%)[e]	VDss (log L/kg)[f]	Fract. Unb.[g]	Total Clearance (log ml/min/kg)[h]
THL	2.89	-2.55	20.23	1.328	93.27	0.049	0.361	1.412

Source: adapted from *SwissADME* and pkCSM (2021).

CONCLUSION

In accordance with the data obtained and support from the literature, we conclude that THL presents anxiolytic and antidepressant-like effects in classic *in vivo* evaluation models, yet without effects on locomotor activity and motor coordination, or other unwanted effects commonly associated with drugs currently used to treat anxiety and depression. As for the molecular docking assays, THL presented satisfactory energy values for nNOs, SGC, IL-6, and 5-HT$_{1A}$, together with superior results for co-crystallized patterns of NMDA and D1 receptors, suggesting their participation THL's mechanism of action. Yet we emphasize that further *in vivo* and *in vitro* tests are needed to confirm or rule out this hypothesis. Finally, the pharmacokinetic models used did not predict or identify potential ADME failures that could limit later preclinical studies. These pioneering studies support THL as a potentially strong multi-target drug candidate - treatment option for the comorbidities of depression and anxiety.

LIST OF ABBREVIATIONS

5-HT = 5-hidroxitriptamine
ANOVA = Analysis of Variance
BDNF = Brain-derived neurotrophic factor
cAMP = Cyclic adenosine monophosphate
cGMP = Cyclic guanosine monophosphate
CNS = Central Nervous System
CREB = cAMP-response element binding protein
DZP = Diazepam
EO = Essential Oil
EPM = Elevated Plus Maze
ERK = Extracellular Signal-Regulated Kinases
FST = Forced Swim Test
GABA = Gamma-Aminobutyric Acid
GAD = Generalized Anxiety Disorder
GC = Guanylyl cyclase
GMQE = Global Model Quality Estimation
GPCR = G-protein-coupled receptors
GTP = Guanosine-5'-triphosphate
HPA = Hypothalamic-pituitary-adrenal axis
IL = Interleukin
LTP = Long-term potentiation
MAOi = Monoamine oxidase inhibitors
MAPK = Mitogen-activated protein kinase
MDD = Major Depressive Disorder
NEOA = Number of entries in the open arms
NET = Norepinephrine transporter
NF-κB = Nuclear factor kappa-light-chain-enhancer of activated B cells
NMDA = N-methyl D-aspartate
NMR = Nucleus medium Rafe
NO = Nitric oxide
NOs = Nitric oxide synthase
OECD = Organisation for Economic Co-operation and Development
PDB = Protein Data Bank
PKA = Protein Kinase A
PSVS = Protein Structure Validation Server
QMEN = Qualitative Model Energy Analysis
RMSD = Root-Mean-Square Deviation
RT = Rotarod Test
SERT = Serotonin transporter
sGC = Soluble guanylate cyclase
SNRIs = Serotonin and Norepinephrine Reuptake Inhibitors
SSRIs: = Selective Serotonin Reuptake Inhibitors
TCA = Tricyclic antidepressant
THL = Tetrahydrolinalool
TNF-α = Tumour Necrosis Factor alpha
TPSA = Topological Polar Surface Area
TrkB = Tropomyosin Receptor Kinase B
VDss = Volume of Distribution at Steady State

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

All experimental procedures were previously approved by the CEUA - Ethics Committee on the Use of Animals at UFPB, under the certificate No. 6479250620.

HUMAN AND ANIMAL RIGHTS

No humans were involved in this study. The reported experiments on animals were in accordance with the Ethics

Committee on the Use of Animals (CEUA) at of the Universidade Federal da Paraíba (UFPB), Brazil.

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

Not applicable.

FUNDING

The Coordination for the Improvement of Higher Education Personnel (Capes) - Brazil - Finance Code 001, and the National Council for Scientific and Technological Development (CNPq) for granting scholarships and to the Federal University of Paraíba (UFPB).

CONFLICT OF INTEREST

The authors declare no conflicts of interest. Sponsors/founders had no role in the study design; in the collection, analysis, or interpretation of data; writing of the manuscript, or the decision to publish the results.

ACKNOWLEDGEMENTS

Declared none.

SUPPLEMENTARY MATERIAL

Supplementary material is available on the publisher's website along with the published article.

REFERENCES

[1] Lord, K.A.; Jacobson, N.C.; Suvak, M.K.; Newman, M.G. Social criticism moderates the relationship between anxiety and depression 10 years later. *J. Affect. Disord.,* **2020**, *274*, 15-22. http://dx.doi.org/10.1016/j.jad.2020.05.030 PMID: 32469798

[2] Wu, Z.; Fang, Y. Comorbidity of depressive and anxiety disorders: Challenges in diagnosis and assessment. *Shanghai Jingshen Yixue,* **2014**, *26*(4), 227-231. PMID: 25317009

[3] Zhou, Y.; Cao, Z.; Yang, M.; Xi, X.; Guo, Y.; Fang, M.; Cheng, L.; Du, Y. Comorbid generalized anxiety disorder and its association with quality of life in patients with major depressive disorder. *Sci. Rep.,* **2017**, *7*(1), 40511. http://dx.doi.org/10.1038/srep40511 PMID: 28098176

[4] Girish, C.; Raj, V.; Arya, J.; Balakrishnan, S. Evidence for the involvement of the monoaminergic system, but not the opioid system in the antidepressant-like activity of ellagic acid in mice. *Eur. J. Pharmacol.,* **2012**, *682*(1-3), 118-125. http://dx.doi.org/10.1016/j.ejphar.2012.02.034 PMID: 22387858

[5] Gomes, P.B.; Feitosa, M.L.; Silva, M.I.G.; Noronha, E.C.; Moura, B.A.; Venâncio, E.T.; Rios, E.R.V.; de Sousa, D.P.; de Vasconcelos, S.M.M.; Fonteles, M.M. de F.; de Sousa, F.C.F. Anxiolytic-like effect of the monoterpene 1,4-cineole in mice. *Pharmacol. Biochem. Behav.,* **2010**, *96*(3), 287-293. http://dx.doi.org/10.1016/j.pbb.2010.05.019 PMID: 20670917

[6] Diniz, T.C.; de Oliveira Júnior, R.G.; Miranda Bezerra Medeiros, M.A.; Gama E Silva, M.; de Andrade Teles, R.B.; Dos Passos Menezes, P.; de Sousa, B.M.H.; Abrahão Frank, L.; de Souza Araújo, A.A.; Russo Serafini, M.; Stanisçuaski Guterres, S.; Pereira Nunes, C.E.; Salvador, M.J.; da Silva Almeida, J.R.G. Anticonvulsant, sedative, anxiolytic and antidepressant activities of the essential oil of Annona vepretorum in mice: Involvement of GABAergic and serotonergic systems. *Biomed. Pharmacother.,* **2019**, *111*, 1074-1087. http://dx.doi.org/10.1016/j.biopha.2018.12.114 PMID: 30841421

[7] Manayi, A.; Nabavi, S.M.; Daglia, M.; Jafari, S. Natural terpenoids as a promising source for modulation of GABAergic system and treatment of neurological diseases. *Pharmacol. Rep.,* **2016**, *68*(4), 671-679. http://dx.doi.org/10.1016/j.pharep.2016.03.014 PMID: 27110875

[8] Santos, P.L.; Matos, J.P.S.C.F.; Picot, L.; Almeida, J.R.G.S.; Quintans, J.S.S.; Quintans-Júnior, L.J. Citronellol, a monoterpene alcohol with promising pharmacological activities - a systematic review. *Food Chem. Toxicol*., **2019**, *123*, 459-469.

[9] Bianchini, A.E.; Garlet, Q.I.; da Cunha, J.A.; Bandeira, G., Junior; Brusque, I.C.M.; Salbego, J.; Heinzmann, B.M.; Baldisserotto, B. Monoterpenoids (thymol, carvacrol and S-(+)-linalool) with anesthetic activity in silver catfish (*Rhamdia quelen*): Evaluation of acetylcholinesterase and GABAergic activity. *Braz. J. Med. Biol. Res.,* **2017**, *50*(12), e6346. http://dx.doi.org/10.1590/1414-431x20176346 PMID: 29069225

[10] Dos Santos, É.R.Q.; Maia, C.S.F.; Fontes Junior, E.A.; Melo, A.S.; Pinheiro, B.G.; Maia, J.G.S. Linalool-rich essential oils from the Amazon display antidepressant-type effect in rodents. *J. Ethnopharmacol.,* **2018**, *212*, 43-49. http://dx.doi.org/10.1016/j.jep.2017.10.013 PMID: 29037915

[11] Stashenko, E.E. Review sampling flower scent for chromatographic analysis. *J. Sep. Sci*., **2008**, *31*(11), 2022-2031.

[12] Harada, H.; Kashiwadani, H.; Kanmura, Y.; Kuwaki, T. Linalool odor-induced anxiolytic effects in mice. *Front. Behav. Neurosci.,* **2018**, *12*, 241. http://dx.doi.org/10.3389/fnbeh.2018.00241 PMID: 30405369

[13] Cheng, B.H.; Sheen, L.Y.; Chang, S.T. Evaluation of anxiolytic potency of essential oil and S-(+)-linalool from *Cinnamomum osmophloeum* ct. linalool leaves in mice. *J. Tradit. Complement. Med.,* **2014**, *5*(1), 27-34. http://dx.doi.org/10.1016/j.jtcme.2014.10.007 PMID: 26151006

[14] Belsito, D.; Bickers, D.; Bruze, M.; Calow, P.; Greim, H.; Hanifin, J.M.; Rogers, A.E.; Saurat, J.H.; Sipes, I.G.; Tagami, H. The RIFM Expert Panel. A safety assessment of non-cyclic alcohols with unsaturated branched chain when used as fragrance ingredients: The RIFM expert panel. *Food Chem. Toxicol.,* **2010**, *48*(Suppl. 3), S1-S42. PMID: 20141871

[15] Kaufman, J.; DeLorenzo, C.; Choudhury, S.; Parsey, R.V. The 5-HT1A receptor in major depressive disorder. *Eur. Neuropsychopharmacol.,* **2016**, *26*(3), 397-410. http://dx.doi.org/10.1016/j.euroneuro.2015.12.039 PMID: 26851834

[16] Derangeon, M.; Bozon, V.; Defamie, N.; Peineau, N.; Bourmeyster, N.; Sarrouilhe, D.; Argibay, J.A.; Hervé, J.C. 5-HT4 and 5-HT2 receptors antagonistically influence gap junctional coupling between rat auricular myocytes. *J. Mol. Cell. Cardiol.,* **2010**, *48*(1), 220-229. http://dx.doi.org/10.1016/j.yjmcc.2009.07.005 PMID: 19615378

[17] Richardson-Jones, J.W.; Craige, C.P.; Nguyen, T.H.; Kung, H.F.; Gardier, A.M.; Dranovsky, A.; David, D.J.; Guiard, B.P.; Beck, S.G.; Hen, R.; Leonardo, E.D. Serotonin-1A autoreceptors are necessary and sufficient for the normal formation of circuits underlying innate anxiety. *J. Neurosci.,* **2011**, *31*(16), 6008-6018. http://dx.doi.org/10.1523/JNEUROSCI.5836-10.2011 PMID: 21508226

[18] Richardson-Jones, J.W.; Craige, C.P.; Guiard, B.P.; Stephen, A.; Metzger, K.L.; Kung, H.F.; Gardier, A.M.; Dranovsky, A.; David, D.J.; Beck, S.G.; Hen, R.; Leonardo, E.D. 5-HT1A autoreceptor levels determine vulnerability to stress and response to antidepressants. *Neuron,* **2010**, *65*(1), 40-52. http://dx.doi.org/10.1016/j.neuron.2009.12.003 PMID: 20152112

[19] Garcia-Garcia, A.L.; Newman-Tancredi, A.; Leonardo, E.D. P5-HT1A Receptors in mood and anxiety: Recent insights into autoreceptor *versus* heteroreceptor function. *Psychopharmacology (Berl.),* **2014**, *231*(4), 623-636. http://dx.doi.org/10.1007/s00213-013-3389-x PMID: 24337875

[20] Rombolà, L.; Scuteri, D.; Watanabe, C.; Sakurada, S.; Hamamura, K.; Sakurada, T.; Tonin, P.; Corasaniti, M.T.; Bagetta, G.; Morrone, L.A. Role of 5-HT1A receptor in the anxiolytic-relaxant ef-

fects of bergamot essential oil in rodent. *Int. J. Mol. Sci.,* **2020**, *21*(7), 1-14.
http://dx.doi.org/10.3390/ijms21072597 PMID: 32283606

[21] Spiacci, A., Jr; Pobbe, R.L.H.; Matthiesen, M.; Zangrossi, H., Jr 5-HT1A receptors of the rat dorsal raphe lateral wings and dorsomedial subnuclei differentially control anxiety- and panic-related defensive responses. *Neuropharmacology,* **2016**, *107*, 471-479.
http://dx.doi.org/10.1016/j.neuropharm.2015.06.015 PMID: 26145183

[22] File, S.E.; Gonzalez, L.E.; Andrews, N. Comparative study of pre- and postsynaptic 5-HT1A receptor modulation of anxiety in two ethological animal tests. *J. Neurosci.,* **1996**, *16*(15), 4810-4815.
http://dx.doi.org/10.1523/JNEUROSCI.16-15-04810.1996 PMID: 8764667

[23] Yohn, C.N.; Gergues, M.M.; Samuels, B.A. The role of 5-HT receptors in depression tim bliss. *Mol. Brain,* **2017**, *10*(1), 1-12.
http://dx.doi.org/10.1186/s13041-017-0306-y

[24] Stiedl, O.; Pappa, E.; Konradsson-Geuken, Å.; Ögren, S.O. The role of the serotonin receptor subtypes 5-HT1A and 5-HT7 and its interaction in emotional learning and memory. *Front. Pharmacol.,* **2015**, *6*, 162.
http://dx.doi.org/10.3389/fphar.2015.00162 PMID: 26300776

[25] Yaman, B.; Bal, R. Pindolol potentiates the antidepressant effect of venlafaxine by inhibiting 5-HT1A receptor in DRN neurons of mice. *Int. J. Neurosci.,* **2022**, *132*, 23-30.
PMID: 32677492

[26] Guzmán, S.L.; Reyes, R.; Bonilla, H. Medicinal plants for the treatment of " Nervios ", anxiety, and depression in mexican traditional medicine. *Rev. Bras. Farmacogn.,* **2014**, *24*(5), 591-608.
http://dx.doi.org/10.1016/j.bjp.2014.10.007

[27] Guzmán-Gutiérrez, S.L.; Bonilla-Jaime, H.; Gómez-Cansino, R.; Reyes-Chilpa, R. Linalool and β-pinene exert their antidepressant-like activity through the monoaminergic pathway. *Life Sci.,* **2015**, *128*, 24-29.
http://dx.doi.org/10.1016/j.lfs.2015.02.021 PMID: 25771248

[28] Ursano, R.J.; Zhang, L.; Li, H.; Johnson, L.; Carlton, J.; Fullerton, C.S.; Benedek, D.M. PTSD and traumatic stress from gene to community and bench to bedside. *Brain Res.,* **2009**, *1293*, 2-12.
http://dx.doi.org/10.1016/j.brainres.2009.03.030 PMID: 19328776

[29] Wang, S.H.; Zhang, Z.J.; Guo, Y.J.; Teng, G.J.; Chen, B.A. Decreased expression of serotonin 1A receptor in the dentate gyrus in association with chronic mild stress: A rat model of post-stroke depression. *Psychiatry Res.,* **2009**, *170*(2-3), 245-251.
http://dx.doi.org/10.1016/j.psychres.2008.07.006 PMID: 19896211

[30] Dantsuji, M.; Nakamura, S.; Nakayama, K.; Mochizuki, A.; Park, S.K.; Bae, Y.C.; Ozeki, M.; Inoue, T. 5-HT_{2A} receptor activation enhances NMDA receptor-mediated glutamate responses through Src kinase in the dendrites of rat jaw-closing motoneurons. *J. Physiol.,* **2019**, *597*(9), 2565-2589.
http://dx.doi.org/10.1113/JP275440 PMID: 30919966

[31] Xiang, M.; Jiang, Y.; Hu, Z.; Yang, Y.; Du, X.; Botchway, B.O.; Fang, M. Serotonin receptors 2A and 1A modulate anxiety-like behavior in post-traumatic stress disordered mice. *Am. J. Transl. Res.,* **2019**, *11*(4), 2288-2303.
PMID: 31105836

[32] Ostrowska, K.; Grzeszczuk, D.; Głuch-Lutwin, M.; Gryboś, A.; Siwek, A.; Leśniak, A.; Sacharczuk, M.; Trzaskowski, B. 5-HT_{1A} and 5-HT_{2A} receptors affinity, docking studies and pharmacological evaluation of a series of 8-acetyl-7-hydroxy-4-methylcoumarin derivatives. *Bioorg. Med. Chem.,* **2018**, *26*(2), 527-535.
http://dx.doi.org/10.1016/j.bmc.2017.12.016 PMID: 29269256

[33] Tiger, M.; Varnäs, K.; Okubo, Y.; Lundberg, J. The 5-HT_{1B} receptor - a potential target for antidepressant treatment. *Psychopharmacology (Berl.),* **2018**, *235*(5), 1317-1334.
http://dx.doi.org/10.1007/s00213-018-4872-1 PMID: 29546551

[34] Nautiyal, K.M.; Tritschler, L.; Ahmari, S.E.; David, D.J.; Gardier, A.M.; Hen, R. A lack of Serotonin 1B autoreceptors results in decreased anxiety and depression-related behaviors. *Neuropsychopharmacology,* **2016**, *41*(12), 2941-2950.
http://dx.doi.org/10.1038/npp.2016.109 PMID: 27353308

[35] Peddie, C.J.; Davies, H.A.; Colyer, F.M.; Stewart, M.G.; Rodríguez, J.J. A subpopulation of serotonin 1B receptors colocalize with the AMPA receptor subunit GluR2 in the hippocampal dentate gyrus. *Neurosci. Lett.,* **2010**, *485*(3), 251-255.
http://dx.doi.org/10.1016/j.neulet.2010.09.024 PMID: 20849926

[36] Veldman, E.R.; Svedberg, M.M.; Svenningsson, P.; Lundberg, J. Distribution and levels of 5-HT_{1B} receptors in anterior cingulate cortex of patients with bipolar disorder, major depressive disorder and schizophrenia - An autoradiography study. *Eur. Neuropsychopharmacol.,* **2017**, *27*(5), 504-514.
http://dx.doi.org/10.1016/j.euroneuro.2017.02.011 PMID: 28318898

[37] Carreño Gutiérrez, H.; O'Leary, A.; Freudenberg, F.; Fedele, G.; Wilkinson, R.; Markham, E.; van Eeden, F.; Reif, A.; Norton, W.H.J. Nitric oxide interacts with monoamine oxidase to modulate aggression and anxiety-like behaviour. *Eur. Neuropsychopharmacol.,* **2020**, *30*, 30-43.
http://dx.doi.org/10.1016/j.euroneuro.2017.09.004 PMID: 28951000

[38] Joca, S.R.L.; Sartim, A.G.; Roncalho, A.L.; Diniz, C.F.A.; Wegener, G. Nitric oxide signalling and antidepressant action revisited. *Cell Tissue Res.,* **2019**, *377*(1), 45-58.
http://dx.doi.org/10.1007/s00441-018-02987-4 PMID: 30649612

[39] Calixto, A.V.; Duarte, F.S.; Duzzioni, M.; Nascimento Häckl, L.P.; Faria, M.S.; De Lima, T.C.M. Role of ventral hippocampal nitric oxide/cGMP pathway in anxiety-related behaviors in rats submitted to the elevated T-maze. *Behav. Brain Res.,* **2010**, *207*(1), 112-117.
http://dx.doi.org/10.1016/j.bbr.2009.09.037 PMID: 19800925

[40] Alderton, W.K.; Cooper, C.E.; Knowles, R.G. Nitric oxide synthases: Structure, function and inhibition. *Biochem. J.,* **2001**, *357*(Pt 3), 593-615.
http://dx.doi.org/10.1042/bj3570593 PMID: 11463332

[41] Lee, C.H.; Giuliani, F. The role of inflammation in depression and fatigue. *Front. Immunol.,* **2019**, *10*, 1696.
http://dx.doi.org/10.3389/fimmu.2019.01696 PMID: 31379879

[42] Pitsikas, N. The role of nitric oxide (NO) donors in anxiety. Lights and shadows. *Nitric Oxide,* **2018**, *77*, 6-11.
http://dx.doi.org/10.1016/j.niox.2018.04.002 PMID: 29625171

[43] Yuste, J.E.; Tarragon, E.; Campuzano, C.M.; Ros-Bernal, F. Implications of glial nitric oxide in neurodegenerative diseases. *Front. Cell. Neurosci.,* **2015**, *9*, 322.
http://dx.doi.org/10.3389/fncel.2015.00322 PMID: 26347610

[44] Salim, S.; Chugh, G.; Asghar, M. *Inflammation in Anxiety,* 1st ed.; Elsevier Inc., **2012**, Vol. 88, .

[45] Coleman, J.W. Nitric oxide in immunity and inflammation. *Int. Immunopharmacol.,* **2001**, *1*(8), 1397-1406.
http://dx.doi.org/10.1016/S1567-5769(01)00086-8 PMID: 11515807

[46] Lau, A.; Tymianski, M. Glutamate receptors, neurotoxicity and neurodegeneration. *Pflugers Arch.,* **2010**, *460*(2), 525-542.
http://dx.doi.org/10.1007/s00424-010-0809-1 PMID: 20229265

[47] Hogg, N.; Kalyanaraman, B. Nitric oxide and lipid peroxidation. *Biochim. Biophys. Acta,* **1999**, *1411*(2-3), 378-384.
http://dx.doi.org/10.1016/S0005-2728(99)00027-4 PMID: 10320670

[48] Denninger, J.W.; Marletta, M.A. Guanylate cyclase and the. NO/cGMP signaling pathway. *Biochim. Biophys. Acta,* **1999**, *1411*(2-3), 334-350.
http://dx.doi.org/10.1016/S0005-2728(99)00024-9 PMID: 10320667

[49] Bal-Price, A.; Brown, G.C. Inflammatory neurodegeneration mediated by nitric oxide from activated glia-inhibiting neuronal respiration, causing glutamate release and excitotoxicity. *J. Neurosci.,* **2001**, *21*(17), 6480-6491.
http://dx.doi.org/10.1523/JNEUROSCI.21-17-06480.2001 PMID: 11517237

[50] Singh, P.; Walia, V. Anxiolytic like effect of l-carnitine in mice: Evidences for the involvement of NO-SGC-CGMP signaling pathway. *Behav. Brain Res*., 2**2020**, *391*, 112689.

[51] Yazir, Y.; Utkan, T.; Aricioglu, F. Inhibition of neuronal nitric oxide synthase and soluble guanylate cyclase prevents depression-like behaviour in rats exposed to chronic unpredictable mild stress. *Basic Clin. Pharmacol. Toxicol.,* **2012**, *111*(3), 154-160. http://dx.doi.org/10.1111/j.1742-7843.2012.00877.x PMID: 22385503

[52] Moretti, M.; Freitas, A.E.; Budni, J.; Fernandes, S.C.P.; Balen, G. de O.; Rodrigues, A.L.S. Involvement of nitric oxide-cGMP pathway in the antidepressant-like effect of ascorbic acid in the tail suspension test. *Behav. Brain Res.,* **2011**, *225*(1), 328-333. http://dx.doi.org/10.1016/j.bbr.2011.07.024 PMID: 21802450

[53] Chaudhari, U.P.; Trivedi, N.D.; Patil, S.R.R.; Banerjee, S. Molecular docking studies of L-Name with the neuronal nitric oxide synthase. *Int. J. Chemtech Res.,* **2010**, *2*, 122-128.

[54] da Silva Leal, V.M.; Bonassoli, V.T.; Soares, L.M.; Milani, H.; de Oliveira, R.M.W. Depletion of 5 hydroxy-triptamine (5-HT) affects the antidepressant-like effect of neuronal nitric oxide synthase inhibitor in mice. *Neurosci. Lett.,* **2017**, *656*, 131-137. http://dx.doi.org/10.1016/j.neulet.2017.07.035 PMID: 28746839

[55] Peana, A.T.; Marzocco, S.; Popolo, A.; Pinto, A. (-)-Linalool inhibits *in vitro* NO formation: Probable involvement in the antinociceptive activity of this monoterpene compound. *Life Sci.,* **2006**, *78*(7), 719-723. http://dx.doi.org/10.1016/j.lfs.2005.05.065 PMID: 16137709

[56] Vogelzangs, N.; de Jonge, P.; Smit, J.H.; Bahn, S.; Penninx, B.W. Cytokine production capacity in depression and anxiety. *Transl. Psychiatry,* **2016**, *6*(5), e825. http://dx.doi.org/10.1038/tp.2016.92 PMID: 27244234

[57] Zou, W.; Feng, R.; Yang, Y. Changes in the serum levels of inflammatory cytokines in antidepressant drug-naïve patients with major depression. *PLoS One,* **2018**, *13*(6), e0197267. http://dx.doi.org/10.1371/journal.pone.0197267 PMID: 29856741

[58] Brymer, K.J.; Romay-Tallon, R.; Allen, J.; Caruncho, H.J.; Kalynchuk, L.E. Exploring the potential antidepressant mechanisms of TNFα antagonists. *Front. Neurosci.,* **2019**, *13*, 98. http://dx.doi.org/10.3389/fnins.2019.00098 PMID: 30804748

[59] Kenis, G.; Maes, M. Effects of antidepressants on the production of cytokines. *Int. J. Neuropsychopharmacol.,* **2002**, *5*(4), 401-412. http://dx.doi.org/10.1017/S1461145702003164 PMID: 12466038

[60] Herr, N.; Bode, C.; Duerschmied, D. The effects of serotonin in immune cells. *Front. Cardiovasc. Med.,* **2017**, *4*, 48. http://dx.doi.org/10.3389/fcvm.2017.00048 PMID: 28775986

[61] Postal, M.; Appenzeller, S. The role of tumor necrosis factor-alpha (TNF-α) in the pathogenesis of systemic lupus erythematosus. *Cytokine,* **2011**, *56*(3), 537-543. http://dx.doi.org/10.1016/j.cyto.2011.08.026 PMID: 21907587

[62] Postal, M.; Lapa, A.T.; Sinicato, N.A.; de Oliveira Peliçari, K.; Peres, F.A.; Costallat, L.T.; Fernandes, P.T.; Marini, R.; Appenzeller, S.; Appenzeller, S. Depressive symptoms are associated with tumor necrosis factor alpha in systemic lupus erythematosus. *J. Neuroinflammation,* **2016**, *13*(1), 5. http://dx.doi.org/10.1186/s12974-015-0471-9 PMID: 26732584

[63] Krishnadas, R.; Cavanagh, J. Depression : An inflammatory illness ?. *J. Neurol. Neurosurg. Psychiatry,* **2012**, *83*(5), 495-502.

[64] Baud, V.; Karin, M.; Karin, M. Signal transduction by tumor necrosis factor and its relatives. *Trends Cell Biol.,* **2001**, *11*, 372-377.

[65] Alshammari, M.A.; Khan, M.R.; Majid Mahmood, H.; Alshehri, A.O.; Alasmari, F.F.; Alqahtani, F.M.; Alasmari, A.F.; Alsharari, S.D.; Alhossan, A.; Ahmad, S.F.; Nadeem, A.; Alshammari, T.K. Systemic TNF-α blockade attenuates anxiety and depressive-like behaviors in *db/db* mice through downregulation of inflammatory signaling in peripheral immune cells. *Saudi Pharm. J.,* **2020**, *28*(5), 621-629. http://dx.doi.org/10.1016/j.jsps.2020.04.001 PMID: 32435144

[66] Uzzan, S.; Azab, A.N. Anti-TNF- α compounds as a treatment for depression. *Molecules*. **2021**, *26*(8): 2368.

[67] Ting, Y-C.; Yang, A.C.; Tsai, S-J. Molecular sciences role of interleukin-6 in depressive disorder. *Int. J. Mol. Sci.,* **2020**, *21*(6), 1-22. http://dx.doi.org/10.3390/ijms21062194

[68] Jones, M.E.; Lebonville, C.L.; Barrus, D.; Lysle, D.T. The role of brain interleukin-1 in stress-enhanced fear learning. *Neuropsychopharmacology,* **2015**, *40*(5), 1289-1296. http://dx.doi.org/10.1038/npp.2014.317 PMID: 25430780

[69] Rossi, S.; Sacchetti, L.; Napolitano, F.; De Chiara, V.; Motta, C.; Studer, V.; Musella, A.; Barbieri, F.; Bari, M.; Bernardi, G.; Maccarrone, M.; Usiello, A.; Centonze, D. Interleukin-1β causes anxiety by interacting with the endocannabinoid system. *J. Neurosci.,* **2012**, *32*(40), 13896-13905. http://dx.doi.org/10.1523/JNEUROSCI.1515-12.2012 PMID: 23035099

[70] Rangel, M.P.; de Mello, J.C.P.; Audi, E.A. Evaluation of neurotransmitters involved in the anxiolytic and panicolytic effect of the aqueous fraction of *Paullinia cupana* (guaraná) in elevated T maze. *Rev. Bras. Farmacogn.,* **2013**, *23*(2), 358-365. http://dx.doi.org/10.1590/S0102-695X2013005000024

[71] Hansen, K.B.; Yi, F.; Perszyk, R.E.; Furukawa, H.; Wollmuth, L.P.; Gibb, A.J.; Traynelis, S.F. Structure, function, and allosteric modulation of NMDA receptors. *J. Gen. Physiol.,* **2018**, *150*(8), 1081-1105. http://dx.doi.org/10.1085/jgp.201812032 PMID: 30037851

[72] Lee, C.H.; Lü, W.; Michel, J.C.; Goehring, A.; Du, J.; Song, X.; Gouaux, E. NMDA receptor structures reveal subunit arrangement and pore architecture. *Nature,* **2014**, *511*(7508), 191-197. http://dx.doi.org/10.1038/nature13548 PMID: 25008524

[73] Niciu, M.J.; Henter, I.D.; Luckenbaugh, D.A.; Zarate, C.A., Jr; Charney, D.S. Glutamate receptor antagonists as fast-acting therapeutic alternatives for the treatment of depression: Ketamine and other compounds. *Annu. Rev. Pharmacol. Toxicol.,* **2014**, *54*(1), 119-139. http://dx.doi.org/10.1146/annurev-pharmtox-011613-135950 PMID: 24392693

[74] Zorumski, C.F.; Izumi, Y.; Mennerick, S. Ketamine: NMDA receptors and beyond. *J. Neurosci.,* **2016**, *36*(44), 11158-11164. http://dx.doi.org/10.1523/JNEUROSCI.1547-16.2016 PMID: 27807158

[75] Barkus, C.; McHugh, S.B.; Sprengel, R.; Seeburg, P.H.; Rawlins, J.N.P.; Bannerman, D.M. Hippocampal NMDA receptors and anxiety: At the interface between cognition and emotion. *Eur. J. Pharmacol.,* **2010**, *626*(1), 49-56. http://dx.doi.org/10.1016/j.ejphar.2009.10.014 PMID: 19836379

[76] Amidfar, M.; Woelfer, M.; Réus, G.Z.; Quevedo, J.; Walter, M.; Kim, Y.K. The role of NMDA receptor in neurobiology and treatment of major depressive disorder: Evidence from translational research. *Prog. Neuropsychopharmacol. Biol. Psychiatry,* **2019**, *94*, 109668. http://dx.doi.org/10.1016/j.pnpbp.2019.109668 PMID: 31207274

[77] Duman, R.S.; Aghajanian, G.K.; Sanacora, G.; Krystal, J.H. Synaptic plasticity and depression: New insights from stress and rapid-acting antidepressants. *Nat. Med.,* **2016**, *22*(3), 238-249. http://dx.doi.org/10.1038/nm.4050 PMID: 26937618

[78] Duman, R.S.; Li, N.; Liu, R.J.; Duric, V.; Aghajanian, G. Signaling pathways underlying the rapid antidepressant actions of ketamine. *Neuropharmacology,* **2012**, *62*(1), 35-41. http://dx.doi.org/10.1016/j.neuropharm.2011.08.044 PMID: 21907221

[79] Duman, R.S.; Li, N. A neurotrophic hypothesis of depression : Role of synaptogenesis in the actions of NMDA receptor antagonists. *Philos. Trans. R Soc. Lond. B Biol. Sci.,* **2016**, *367,* 2475-2484.

[80] López, V.; Nielsen, B.; Solas, M.; Ramírez, M.J.; Jäger, A.K. Exploring pharmacological mechanisms of lavender (*Lavandula angustifolia*) essential oil on central nervous system targets. *Front. Pharmacol.,* **2017**, *8*, 280. http://dx.doi.org/10.3389/fphar.2017.00280 PMID: 28579958

[81] Pourtaqi, N.; Imenshahidi, M.; Razavi, B.M.; Hosseinzadeh, H. Effect of linalool on the acquisition and reinstatement of morphine-induced conditioned place preference in mice. *Avicenna J. Phytomed.,* **2017**, *7*(3), 242-249. PMID: 28748171

[82] Stepanenko, Y.D.; Boikov, S.I.; Sibarov, D.A.; Abushik, P.A.; Vanchakova, N.P.; Belinskaia, D.; Shestakova, N.N.; Antonov,

S.M. Dual action of amitriptyline on NMDA receptors: Enhancement of Ca-dependent desensitization and trapping channel block. *Sci. Rep.,* **2019**, *9*(1), 19454.
http://dx.doi.org/10.1038/s41598-019-56072-z PMID: 31857688

[83] Sibarov, D.A.; Giniatullin, R.; Antonov, S.M. High sensitivity of cerebellar neurons to homocysteine is determined by expression of GluN2C and GluN2D subunits of NMDA receptors. *Biochem. Biophys. Res. Commun.,* **2018**, *506*(3), 648-652.
http://dx.doi.org/10.1016/j.bbrc.2018.10.140 PMID: 30454701

[84] Volkmann, R.A.; Fanger, C.M.; Anderson, D.R.; Sirivolu, V.R.; Paschetto, K.; Gordon, E.; Virginio, C.; Gleyzes, M.; Buisson, B.; Steidl, E.; Mierau, S.B.; Fagiolini, M.; Menniti, F.S. MPX-004 and MPX-007: New pharmacological tools to study the physiology of nmda receptors containing the GluN2A subunit. *PLoS One,* **2016**, *11*(2), e0148129.
http://dx.doi.org/10.1371/journal.pone.0148129 PMID: 26829109

[85] Edman, S.; McKay, S.; Macdonald, L.J.; Samadi, M.; Livesey, M.R.; Hardingham, G.E.; Wyllie, D.J.A. TCN 201 selectively blocks GluN2A-containing NMDARs in a GluN1 co-agonist dependent but non-competitive manner. *Neuropharmacology,* **2012**, *63*(3), 441-449.
http://dx.doi.org/10.1016/j.neuropharm.2012.04.027 PMID: 22579927

[86] Hasbi, A.; Perreault, M.L.; Shen, M.Y.F.; Zhang, L.; To, R.; Fan, T.; Nguyen, T.; Ji, X.; O'Dowd, B.F.; George, S.R. A peptide targeting an interaction interface disrupts the dopamine D1-D2 receptor heteromer to block signaling and function *in vitro* and *in vivo*: Effective selective antagonism. *FASEB J.,* **2014**, *28*(11), 4806-4820.
http://dx.doi.org/10.1096/fj.14-254037 PMID: 25063849

[87] Shen, M.Y.F.; Perreault, M.L.; Bambico, F.R.; Jones-Tabah, J.; Cheung, M.; Fan, T.; Nobrega, J.N.; George, S.R. Rapid antidepressant and anxiolytic actions following dopamine D1-D2 receptor heteromer inactivation. *Eur. Neuropsychopharmacol.,* **2015**, *25*(12), 2437-2448.
http://dx.doi.org/10.1016/j.euroneuro.2015.09.004 PMID: 26431907

[88] Perreault, M.L.; Hasbi, A.; O'Dowd, B.F.; George, S.R. Heteromeric dopamine receptor signaling complexes: Emerging neurobiology and disease relevance. *Neuropsychopharmacology,* **2014**, *39*(1), 156-168.
http://dx.doi.org/10.1038/npp.2013.148 PMID: 23774533

[89] Mańko, M.; Geracitano, R.; Capogna, M. Functional connectivity of the main intercalated nucleus of the mouse amygdala. *J. Physiol.,* **2011**, *589*(Pt 8), 1911-1925.
http://dx.doi.org/10.1113/jphysiol.2010.201475 PMID: 21224220

[90] Pavlova, I.V.; Rysakova, M.P.; Sergeeva, M.I. Effects of blockade of D1 and D2 receptors in the basolateral amygdala on the behavior of rats with high and low levels of anxiety and fear. *Neurosci. Behav. Physiol.,* **2016**, *46*(9), 1-11.
http://dx.doi.org/10.1007/s11055-016-0352-4

[91] Russo, S.J.; Nestler, E.J. The brain reward circuitry in mood disorders. *Nat. Rev. Neurosci.,* **2013**, *14*(9), 609-625.
http://dx.doi.org/10.1038/nrn3381 PMID: 23942470

[92] Duman, R.S.; Voleti, B. Signaling pathways underlying the pathophysiology and treatment of depression: Novel mechanisms for rapid-acting agents. *Trends Neurosci.,* **2012**, *35*(1), 47-56.
http://dx.doi.org/10.1016/j.tins.2011.11.004 PMID: 22217452

[93] Hasbi, A.; Nguyen, T.; Rahal, H.; Manduca, J.D.; Miksys, S.; Tyndale, R.F.; Madras, B.K.; Perreault, M.L.; George, S.R. Sex difference in dopamine D1-D2 receptor complex expression and signaling affects depression- and anxiety-like behaviors. *Biol. Sex Differ.,* **2020**, *11*(1), 8.
http://dx.doi.org/10.1186/s13293-020-00285-9 PMID: 32087746

[94] Gilsbach, R.; Hein, L. Are the pharmacology and physiology of α_2 adrenoceptors determined by α_2-heteroreceptors and autoreceptors respectively? *Br. J. Pharmacol.,* **2012**, *165*(1), 90-102.
http://dx.doi.org/10.1111/j.1476-5381.2011.01533.x PMID: 21658028

[95] Uys, M.M.; Shahid, M.; Harvey, B.H. Therapeutic potential of selectively targeting the α_{2c}-adrenoceptor in cognition, depression, and schizophrenia-new developments and future perspective. *Front. Psychiatry,* **2017**, *8*, 144.
http://dx.doi.org/10.3389/fpsyt.2017.00144 PMID: 28855875

[96] Pertovaara, A. The noradrenergic pain regulation system: A potential target for pain therapy. *Eur. J. Pharmacol.,* **2013**, *716*(1-3), 2-7.
http://dx.doi.org/10.1016/j.ejphar.2013.01.067 PMID: 23500194

[97] Höcker, J.; Weber, B.; Tonner, P.H.; Scholz, J.; Brand, P.A.; Ohnesorge, H.; Bein, B. Meperidine, remifentanil and tramadol but not sufentanil interact with A2-adrenoceptors in A2A-, A2B- and A2C-adrenoceptor knock out mice brain. *Eur. J. Pharmacol.,* **2008**, *582*, 70-77.
http://dx.doi.org/10.1016/j.ejphar.2007.12.022 PMID: 18234187

[98] Millan, M.J. Evidence that an alpha 2A-adrenoceptor subtype mediates antinociception in mice. *Eur. J. Pharmacol.,* **1992**, *215*(2-3), 355-356.
http://dx.doi.org/10.1016/0014-2999(92)90059-D PMID: 1356794

[99] Balogh, B.; Szilágyi, A.; Gyires, K.; Bylund, D.B.; Mátyus, P. Molecular modelling of subtypes (A2A, A2B and A2C) of A2-adrenoceptors: A comparative study. *Neurochem. Int.,* **2009**, *55*, 355-361.
http://dx.doi.org/10.1016/j.neuint.2009.05.004 PMID: 19447153

[100] Ostopovici-Halip, L.; Curpăn, R.; Mracec, M.; Bologa, C.G. Structural determinants of the alpha2 adrenoceptor subtype selectivity. *J. Mol. Graph. Model.,* **2011**, *29*(8), 1030-1038.
http://dx.doi.org/10.1016/j.jmgm.2011.04.011 PMID: 21602069

[101] Diamanti, E.; Del Bello, F.; Carbonara, G.; Carrieri, A.; Fracchiolla, G.; Giannella, M.; Mammoli, V.; Piergentili, A.; Pohjanoksa, K.; Quaglia, W.; Scheinin, M.; Pigini, M. Might the observed α(2A)-adrenoreceptor agonism or antagonism of allyphenyline analogues be ascribed to different molecular conformations? *Bioorg. Med. Chem.,* **2012**, *20*(6), 2082-2090.
http://dx.doi.org/10.1016/j.bmc.2012.01.035 PMID: 22341244

[102] Sharp, T.; Cowen, P.J. 5-HT and depression: Is the glass half-full? *Curr. Opin. Pharmacol.,* **2011**, *11*(1), 45-51.
http://dx.doi.org/10.1016/j.coph.2011.02.003 PMID: 21377932

[103] Action, A.; Serotonin, I.S.; Relevant, S.; An, T.O.; Depression, U.O.F. What has serotonin to do with depression. *World Psychiatry*, **2015**, *14*(2), 158-160.

[104] Moret, C. Importance of norepinephrine in depression. *Neuropsychiatr. Dis. Treat.,* **2011**, *7*, 9-13.

[105] Sørensen, L.; Andersen, J.; Thomsen, M.; Hansen, S.M.R.; Zhao, X.; Sandelin, A.; Strømgaard, K.; Kristensen, A.S. Interaction of antidepressants with the serotonin and norepinephrine transporters. *J. Biol. Chem.,* **2012**, *287*, 43694-43707.

[106] Andersen, J.; Taboureau, O.; Hansen, K.B.; Olsen, L.; Egebjerg, J.; Strømgaard, K.; Kristensen, A.S. Location of the antidepressant binding site in the serotonin transporter importance of SER-438 in recognition of citalopram and tricyclic. *J. Biol. Chem*., **2009**, *284*, 10276-10284.

[107] Sinning, S.; Musgaard, M.; Jensen, M.; Severinsen, K.; Celik, L.; Koldsø, H.; Meyer, T.; Bols, M.; Jensen, H.H.; Schiøtt, B.; Wiborg, O. Binding and orientation of tricyclic antidepressants within the central substrate site of the human serotonin. *JBC*, **2010**, *285*, 8363-8374.
http://dx.doi.org/10.1074/jbc.M109.045401

[108] Pellow, S.; File, S.E. Anxiolytic and anxiogenic drug effects on exploratory activity in an elevated plus-maze: A novel test of anxiety in the rat. *Pharmacol. Biochem. Behav.,* **1986**, *24*(3), 525-529.
http://dx.doi.org/10.1016/0091-3057(86)90552-6 PMID: 2871560

[109] Archer, J. Tests for emotionality in rats and mice: A review. *Anim. Behav.,* **1973**, *21*(2), 205-235.
http://dx.doi.org/10.1016/S0003-3472(73)80065-X PMID: 4578750

[110] Pultrini, A.M.; Galindo, L.A.; Costa, M. Effects of the essential oil from Citrus aurantium L. in experimental anxiety models in mice. *Life Sci.,* **2006**, *78*(15), 1720-1725.
http://dx.doi.org/10.1016/j.lfs.2005.08.004 PMID: 16253279

[111] Dunham, N.W.; Miya, T.S. A note on a simple apparatus for detecting neurological deficit in rats and mice. *J. Am. Pharm. Assoc. Am. Pharm. Assoc. (Baltim),* **1957**, *46*, 208-209.

http://dx.doi.org/10.1002/jps.3030460322

[112] Can, A.; Dao, D.T.; Arad, M.; Terrillion, C.E.; Piantadosi, S.C.; Gould, T.D. The mouse forced swim test. *J. Vis. Exp.,* **2012**, *59*, e3638.
PMID: 22314943

[113] Bernstein, F.C.; Koetzle, T.F.; Williams, G.J.B.; Meyer, E.F., Jr; Brice, M.D.; Rodgers, J.R.; Kennard, O.; Shimanouchi, T.; Tasumi, M. The protein data bank: A computer-based archival file for macromolecular structures. *Arch. Biochem. Biophys.,* **1978**, *185*(2), 584-591.
http://dx.doi.org/10.1016/0003-9861(78)90204-7 PMID: 626512

[114] Waterhouse, A.; Bertoni, M.; Bienert, S.; Studer, G.; Tauriello, G.; Gumienny, R.; Heer, F.T.; de Beer, T.A.P.; Rempfer, C.; Bordoli, L.; Lepore, R.; Schwede, T. SWISS-MODEL: Homology modelling of protein structures and complexes. *Nucleic Acids Res.,* **2018**, *46*(W1), W296-W303.
http://dx.doi.org/10.1093/nar/gky427 PMID: 29788355

[115] Laskowski, R.A.; MacArthur, M.W.; Moss, D.S.; Thornton, J.M. PROCHECK: A program to check the stereochemical quality of protein structures. *J. Appl. Cryst.,* **1993**, *26*(2), 283-291.
http://dx.doi.org/10.1107/S0021889892009944

[116] Lovell, S.C.; Davis, I.W.; Adrendall, W.B.; de Bakker, P.I.W.; Word, J.M.; Prisant, M.G.; Richardson, J.S.; Richardson, D.C. Structure validation by Calpha geometry: phi,psi and Cbeta deviation. *Proteins.* **2003**, *50*(3), 437-50.

[117] Pires, D.E.V.; Blundell, T.L.; Ascher, D.B. pkCSM: Predicting small-molecule pharmacokinetic and toxicity properties using graph-based signatures. *J. Med. Chem.,* **2015**, *58*(9), 4066-4072.
http://dx.doi.org/10.1021/acs.jmedchem.5b00104 PMID: 25860834

[118] Daina, A.; Michielin, O.; Zoete, V. SwissADME: A free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Sci. Rep.,* **2017**, *7*(1), 42717.
http://dx.doi.org/10.1038/srep42717 PMID: 28256516

[119] Lapczynski, A.; Foxenberg, R.J.; Bhatia, S.P.; Letizia, C.S.; Api, A.M. Fragrance material review on tetrahydrolinalool. *Food Chem. Toxicol.,* **2008**, *46*(Suppl. 11), S286-S288.
http://dx.doi.org/10.1016/j.fct.2008.06.076 PMID: 18640232

[120] de Almeida, A.A.C.; Costa, J.P.; de Carvalho, R.B.F.; de Sousa, D.P.; de Freitas, R.M. Evaluation of acute toxicity of a natural compound (+)-limonene epoxide and its anxiolytic-like action. *Brain Res.,* **2012**, *1448*, 56-62.
http://dx.doi.org/10.1016/j.brainres.2012.01.070 PMID: 22364736

[121] Guzmán-Gutiérrez, S.L.; Gómez-Cansino, R.; García-Zebadúa, J.C.; Jiménez-Pérez, N.C.; Reyes-Chilpa, R. Antidepressant activity of Litsea glaucescens essential oil: Identification of β-pinene and linalool as active principles. *J. Ethnopharmacol.,* **2012**, *143*(2), 673-679.
http://dx.doi.org/10.1016/j.jep.2012.07.026 PMID: 22867633

[122] Umezu, T.; Nagano, K.; Ito, H.; Kosakai, K.; Sakaniwa, M.; Morita, M. Anticonflict effects of lavender oil and identification of its active constituents. *Pharmacol. Biochem. Behav.,* **2006**, *85*(4), 713-721.
http://dx.doi.org/10.1016/j.pbb.2006.10.026 PMID: 17173962

[123] Mori, T.; Shibasaki, M.; Ogawa, Y.; Hokazono, M.; Wang, T.C.; Rahmadi, M.; Suzuki, T. Comparison of the behavioral effects of bupropion and psychostimulants. *Eur. J. Pharmacol.,* **2013**, *718*(1-3), 370-375.
http://dx.doi.org/10.1016/j.ejphar.2013.07.046 PMID: 23993950

[124] Nielsen, J.A.; Shannon, N.J.; Bero, L.; Moore, K.E. Effects of acute and chronic bupropion on locomotor activity and dopaminergic neurons. *Pharmacol. Biochem. Behav.,* **1986**, *24*(4), 795-799.
http://dx.doi.org/10.1016/0091-3057(86)90413-2 PMID: 3086903

[125] Rauhut, A.S.; Hawrylak, M.; Mardekian, S.K. Bupropion differentially alters the aversive, locomotor and rewarding properties of nicotine in CD-1 mice. *Pharmacol. Biochem. Behav.,* **2008**, *90*(4), 598-607.
http://dx.doi.org/10.1016/j.pbb.2008.05.002 PMID: 18556053

[126] Oil, L. Evaluation of motor coordination and antidepressant activities of *Cinnamomum osmophloeum* Ct. linalool leaf oil in rodent model. *Molecules*, **2021**, *26*(10), 3037.

[127] Souto-Maior, F.N.; Fonsêca, D.V.; Salgado, P.R.R.; Monte, L.O.; de Sousa, D.P.; de Almeida, R.N. Antinociceptive and anticonvulsant effects of the monoterpene linalool oxide. *Pharm. Biol.,* **2017**, *55*(1), 63-67.
http://dx.doi.org/10.1080/13880209.2016.1228682 PMID: 27622736

[128] Lader, M.H. Limitations on the use of benzodiazepines in anxiety and insomnia: Are they justified? *Eur. Neuropsychopharmacol.,* **1999**, *9*(Suppl. 6), S399-S405.
http://dx.doi.org/10.1016/S0924-977X(99)00051-6 PMID: 10622686

[129] Prevot, T.D.; Li, G.; Vidojevic, A.; Misquitta, K.A.; Fee, C.; Santrac, A.; Knutson, D.E.; Stephen, M.R.; Kodali, R.; Zahn, N.M.; Arnold, L.A.; Scholze, P.; Fisher, J.L.; Marković, B.D.; Banasr, M.; Cook, J.M.; Savic, M.; Sibille, E. Novel benzodiazepine-like ligands with various anxiolytic, antidepressant, or pro-cognitive profiles. *Mol. Neuropsychiatry,* **2019**, *5*(2), 84-97.
http://dx.doi.org/10.1159/000496086 PMID: 31192221

[130] Yuen, E.; Swanson, S.; Witkin, J.M. Prediction of human efficacious antidepressant doses using the mouse forced swim test. *Pharmacol. Biochem. Behav.,* **2017**, *161*, 22-29.
http://dx.doi.org/10.1016/j.pbb.2017.09.002 PMID: 28888484

[131] Kalueff, A.V.; Tuohimaa, P. Experimental modeling of anxiety and depression. *Acta Neurobiol. Exp. (Warsz.),* **2004**, *64*(4), 439-448.
PMID: 15586660

[132] Rizvi, S.J.; Pizzagalli, D.A.; Sproule, B.A.; Kennedy, S.H. Assessing anhedonia in depression: Potentials and pitfalls. *Neurosci. Biobehav. Rev.,* **2016**, *65*, 21-35.
http://dx.doi.org/10.1016/j.neubiorev.2016.03.004 PMID: 26959336

[133] McMakin, D.L.; Olino, T.M.; Porta, G.; Dietz, L.J.; Emslie, G.; Clarke, G.; Wagner, K.D.; Asarnow, J.R.; Ryan, N.D.; Birmaher, B.; Shamseddeen, W.; Mayes, T.; Kennard, B.; Spirito, A.; Keller, M.; Lynch, F.L.; Dickerson, J.F.; Brent, D.A. Anhedonia predicts poorer recovery among youth with selective serotonin reuptake inhibitor treatment-resistant depression. *J. Am. Acad. Child Adolesc. Psychiatry,* **2012**, *51*(4), 404-411.
http://dx.doi.org/10.1016/j.jaac.2012.01.011 PMID: 22449646

[134] Uher, R.; Perlis, R.H.; Henigsberg, N.; Zobel, A.; Rietschel, M.; Mors, O.; Hauser, J.; Dernovsek, M.Z.; Souery, D.; Bajs, M.; Maier, W.; Aitchison, K.J.; Farmer, A.; McGuffin, P. Depression symptom dimensions as predictors of antidepressant treatment outcome: Replicable evidence for interest-activity symptoms. *Psychol. Med.,* **2012**, *42*(5), 967-980.
http://dx.doi.org/10.1017/S0033291711001905 PMID: 21929846

[135] Müller, W.E.; Sillani, G.; Schuwald, A.; Friedland, K. Pharmacological basis of the anxiolytic and antidepressant properties of Silexan®, an essential oil from the flowers of lavender. *Neurochem. Int.,* **2021**, *143*, 104899.
http://dx.doi.org/10.1016/j.neuint.2020.104899 PMID: 33181239

[136] Kasper, S.; Gastpar, M.; Müller, W.E.; Volz, H.P.; Möller, H.J.; Dienel, A.; Schläfke, S. Silexan, an orally administered Lavandula oil preparation, is effective in the treatment of ‘subsyndromal’ anxiety disorder: A randomized, double-blind, placebo controlled trial. *Int. Clin. Psychopharmacol.,* **2010**, *25*(5), 277-287.
http://dx.doi.org/10.1097/YIC.0b013e32833b3242 PMID: 20512042

[137] Sartori, D.P.; Oliveira, N.F.; Valentim, J.T.; Silva, D.M.A.; Mallman, A.S.V.; Oliveira, I.C.M.; Chaves, R.C.; Capibaribe, V.C.; Carvalho, A.M.R.; Rebouças, M.O.; Macedo, D.S.; Chaves Filho, A.J.M.; Fonteles, M.M.F.; Gutierrez, S.J.C.; Barbosa-Filho, J.M.; Mottin, M.; Andrade, C.H.; Sousa, F.C.F. Involvement of monoaminergic targets in the antidepressant- and anxiolytic-like effects of the synthetic alkamide riparin IV: Elucidation of further mechanisms through pharmacological, neurochemistry and computational approaches. *Behav. Brain Res.,* **2020**, *383*, 112487.
http://dx.doi.org/10.1016/j.bbr.2020.112487 PMID: 31987932

[138] da Silva Calixto, P.; de Almeida, R.N.; Stiebbe Salvadori, M.G.S.; Dos Santos Maia, M.; Filho, J.M.B.; Scotti, M.T.; Scotti, L. *In silico* study examining new phenylpropanoids targets with antidepressant activity. *Curr. Drug Targets,* **2021**, *22*(5), 539-554.

http://dx.doi.org/10.2174/1389450121666200902171838 PMID: 32881667

[139] Isyaku, Y.; Uzairu, A.; Uba, S. Computational studies of a series of 2-substituted phenyl-2-oxo-, 2-hydroxyl- and 2-acylloxyethylsulfonamides as potent anti-fungal agents. *Heliyon,* **2020**, *6*(4), e03724. http://dx.doi.org/10.1016/j.heliyon.2020.e03724 PMID: 32322718

[140] Wang, Y.; Liu, H.; Fan, Y.; Chen, X.; Yang, Y.; Zhu, L.; Zhao, J.; Chen, Y.; Zhang, Y. *In silico* prediction of human intravenous pharmacokinetic parameters with improved accuracy. *J. Chem. Inf. Model.,* **2019**, *59*(9), 3968-3980. http://dx.doi.org/10.1021/acs.jcim.9b00300 PMID: 31403793

[141] Daina, A.; Zoete, V. A boiled-egg to predict gastrointestinal absorption and brain penetration of small molecules. *ChemMedChem,* **2016**, *11*(11), 1117-1121. http://dx.doi.org/10.1002/cmdc.201600182 PMID: 27218427

[142] Gfeller, D.; Grosdidier, A.; Wirth, M.; Daina, A.; Michielin, O.; Zoete, V. SwissTargetPrediction: A web server for target prediction of bioactive small molecules. *Nucleic Acids Res.,* **2014**, *42*, W32-8. http://dx.doi.org/10.1093/nar/gku293 PMID: 24792161

[143] Mohamed, A.A.; El-Kadi, A. P-Glycoprotein effects on drugs pharmacokinetics and drug-drug- interactions and their clinical implications. *Libyan J. Pharm. Clin. Pharmacol.,* **2012**, *1,* 48154.

[144] Mealey, K.L.; Waiting, D.; Raunig, D.L.; Schmidt, K.R.; Nelson, F.R. Oral bioavailability of P-glycoprotein substrate drugs do not differ between ABCB1-1Δ and ABCB1 wild type dogs. *J. Vet. Pharmacol. Ther.,* **2010**, *33*(5), 453-460. http://dx.doi.org/10.1111/j.1365-2885.2010.01170.x PMID: 20840389

[145] Silva, P.; de Almeida, M.; Silva, J.; Albino, S.; Espírito-Santo, R.; Lima, M.; Villarreal, C.; Moura, R.; Santos, V. (E)-2-Cyano-3-(1*H*-Indol-3-yl)-N-Phenylacrylamide, a hybrid compound derived from indomethacin and paracetamol: Design, synthesis and evaluation of the anti-inflammatory potential. *Int. J. Mol. Sci.,* **2020**, *21*(7), 1-17. http://dx.doi.org/10.3390/ijms21072591 PMID: 32276463

[146] Pires, D.E. V; Blundell, T.L.; Ascher, D.B. PkCSM: Predicting small-molecule pharmacokinetic properties using graph-based signatures (Theory- how to enterpret pkcsm result). *J. Med. Chem.,* **2015***, 58, 9,* 4066-4072.